

INTELLIGENT VEHICLE SPEED LIMITING AND MONITORING SYSTEM USING HYBRID IR-LiFi-LoRa COMMUNICATION

PROJECT REPORT

PHASE – II

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in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

DEPARTMENT OF INFORMATION TECHNOLOGY



MANAKULA VINAYAGAR INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

KALITHEERTHALKUPPAM, PUDUCHERRY- 605 107.

PONDICHERRY UNIVERSITY

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BONAFIDE CERTIFICATE

This is to certify that the Project work titled "INTELLIGENT VEHICLE SPEED LIMITING AND MONITORING SYSTEM USING HYBRID IR-LiFi-LoRa COMMUNICATION" is a bona fide work done by **DHARSHNAMOORTHY R [REG.NO: 22TH0223]**, **GOGULAKANNAN M [REG.NO: 22TH0228]**, **HENRY DANIEL DIDACE D [REG.NO: 22TH0232]** and **PRASHANTH ARJUN GK [REG.NO: 22TH0276]** in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Information Technology by Pondicherry University during the academic year 2025-26.

PROJECT GUIDE

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

This is certified that the Report entitled "**INTELLIGENT VEHICLE SPEED LIMITING AND MONITORING SYSTEM USING HYBRID IR-LiFi-LoRa COMMUNICATION**" is the bona fide record of independent work done by **DHARSHNAMOORTHY R [REG.NO: 22TH0223]**, **GOGULAKANNAN M [REG.NO: 22TH0228]**, **HENRY DANIEL DIDACE D [REG.NO: 22TH0232]** and **PRASHANTH ARJUN GK [REG.NO: 22TH0276]**, for the award of B.Tech. Degree in Department of Information Technology under the supervision of **Mrs. V. VIMALA DHEEKSHANYA**. Certified further that the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred earlier.

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We thank the Almighty for blessing us with such wonderful people and for being with us always.

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DETAILS OF THE PROJECT

Title of the Project	Intelligent Vehicle Speed Limiting and Monitoring System Using Hybrid IR-LiFi-LoRa Communication
Department	Information Technology
Team Members	DHARSHNAMEOORTHY R REG.NO: 22TH0223 GOGULAKANNAN M REG.NO: 22TH0228 HENRY DANIEL DIDACE D REG.NO: 22TH0232 PRASHANTH ARJUN GK REG.NO: 22TH0276
Year / Duration	2025-26 / 6 months
Problem Statement	Road accidents caused by over-speeding remain a critical safety concern, particularly in sensitive zones such as school areas, hospitals, residential streets, and construction sites. Existing speed regulation mechanisms - including speed breakers, static signboards, radar-based surveillance, and GPS-assisted advisory systems — are either static, reactive, or dependent on expensive and continuous network infrastructure. They lack the ability to autonomously enforce speed limits in real time without driver intervention. Furthermore, advanced V2X and cellular-based systems are cost-prohibitive for large-scale deployment in developing regions. There is a clear need for a low-cost, autonomous, and reliable system that can dynamically enforce vehicle speed limits in restricted zones without relying on continuous high-bandwidth connectivity or driver compliance.
Objectives of the Project	This project presents an intelligent vehicular speed control system that leverages a hybrid communication approach combining Infrared (IR), Light Fidelity (LiFi), and Long Range (LoRa) technologies. The system consists of a roadside unit, which detects vehicle presence via an IR sensor and transmits the permissible speed limit using LiFi-based optical communication, while the vehicle unit receives, processes, and enforces the limit through PWM-based motor control, and LoRa is used for telemetry purposes.

	• To design and implement a hybrid IR-LiFi-LoRa communication-based system that automatically enforces vehicle speed limits in restricted zones without driver intervention. • To utilize LiFi-based optical communication for real-time, low-cost speed limit transmission from roadside units to vehicles. • To implement PWM-based motor speed control for smooth and automatic speed regulation upon entering restricted zones. • To enable remote monitoring of vehicle speed and system status via LoRa communication to a monitoring station. • To provide a fail-safe mechanism and manual override feature for safety under signal loss or emergency conditions.
Tools & Platforms	Hardware: • ESP32 Microcontrollers • IR Sensor • High-brightness LED (LiFi Transmitter) • Phototransistor (LiFi Receiver) • L298N Motor Driver • DC Motors (RC Car) • LoRa Module (SX1278) • Push Button (Override) • Buzzer • Battery Pack Software: • Visual Studio Code with PlatformIO, Embedded C / Arduino Framework, LoRa Library (SX1278), PWM / LEDC Functions (ESP32) • Cloud IoT Platform • Mobile Dashboard
PO's / PSO's Mapped	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12 / PSO2

SDGs Mapped	• SDG 3 – Good Health and Well-Being • SDG 9 – Industry, Innovation and Infrastructure • SDG 11 – Sustainable Cities and Communities
SDG Justification (in 100 words)	SDG 3: Good Health and Well-Being: This project directly contributes to SDG 3 by reducing road accidents caused by over-speeding in sensitive zones such as schools and hospitals, thereby protecting human lives. SDG 9: Industry, Innovation and Infrastructure: It aligns with SDG 9 through the development of an innovative, low-cost hybrid communication system combining IR, LiFi, and LoRa technologies for intelligent transportation. SDG 11: Sustainable Cities and Communities: It supports SDG 11 by enabling smarter, safer urban mobility without reliance on expensive infrastructure, making it scalable for smart city environments and developing regions.



SUSTAINABLE DEVELOPMENT GOALS (SDGs) MAPPING

**Title: INTELLIGENT VEHICLE SPEED LIMITING AND MONITORING SYSTEM
USING HYBRID IR-LiFi-LoRa COMMUNICATION**

SDG Goal: SDG 3 - Good Health and Well-Being



SDG Goal: SDG 9 - Industry, Innovation and Infrastructure



SDG Goal: SDG 11 - Sustainable Cities and Communities



SDG Goal - 3: Good Health and Well-Being

This project directly contributes to SDG 3 by targeting a leading cause of road fatalities — over-speeding in sensitive zones such as school areas, hospitals, and residential streets. By automatically enforcing speed limits without depending on driver compliance, the system reduces the likelihood of accidents in high-risk areas. The technology intervenes in real time, preventing collisions before they occur rather than responding after the fact. Through improved road safety, the project helps protect the health and lives of pedestrians, patients, children, and vehicle occupants, directly aligning with the goal of reducing injuries and deaths from road traffic accidents worldwide.

SDG Goal - 9: Industry, Innovation and Infrastructure

The proposed system represents a meaningful innovation in intelligent transportation infrastructure. By combining Infrared (IR) detection, Light Fidelity (LiFi) optical communication, and Long Range (LoRa) wireless telemetry into a unified low-cost architecture, the project advances the state of practice in smart vehicular systems. Unlike existing expensive V2X or 5G-based approaches, this hybrid system offers a scalable and cost-effective alternative that can be deployed in developing regions. It demonstrates how novel combinations of accessible technologies can produce resilient and innovative infrastructure solutions.

SDG Goal - 11: Sustainable Cities and Communities

Smart cities require intelligent, adaptive transportation systems that can enforce safety regulations without burdening citizens with expensive infrastructure. This project contributes to SDG 11 by enabling dynamic, zone-based speed enforcement that can be deployed across schools, hospitals, construction areas, and residential streets with minimal cost and infrastructure requirements. The system operates independently of continuous internet connectivity, making it viable for both urban and semi-urban environments. Its modular design supports scalability across different road types and regions, reinforcing the vision of inclusive, safe, and sustainable urban mobility.

ABSTRACT

This project proposes an Intelligent Vehicle Speed Limiting and Monitoring System that utilizes a hybrid communication approach combining Infrared (IR), Light Fidelity (LiFi), and Long Range (LoRa) technologies. The system is designed to automatically regulate the speed of a robotic vehicle based on predefined restricted speed zones, enhancing road safety and reducing dependence on driver intervention.

The roadside unit detects the presence of a vehicle using an IR sensor and transmits the permissible speed limit using LiFi-based optical communication through a modulated LED signal. The vehicle-mounted unit receives this signal via a photodetector, processes the data, and dynamically adjusts the motor speed using Pulse Width Modulation (PWM) control. This ensures smooth, real-time, and automatic speed regulation without relying on driver response.s

Additionally, the system incorporates LoRa communication for the continuous transmission of vehicle speed and system status to a remote monitoring station, enabling real-time supervision of the vehicle. A manual override mechanism is also implemented, allowing the driver to temporarily bypass speed restrictions for a defined duration in emergency conditions, after which the system automatically resumes normal operation.

The hybrid system ensures low-cost implementation, real-time response, and improved reliability, making it suitable for prototype-level intelligent transportation systems and smart city applications.

By combining localized optical communication with long-range wireless telemetry, the proposed architecture minimizes dependency on expensive cellular or satellite networks while delivering a robust, scalable, and autonomous speed control solution.

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LIST OF ABBREVIATIONS

ABBREVIATIONS	FULL FORM
ABS	Anti-lock Braking System
ADAS	Advanced Driver Assistance Systems
CSS	Chirp Spread Spectrum
ESP32	Espressif 32-bit Microcontroller
IIoT	Industrial Internet of Things
IoT	Internet of Things
IR	Infrared
ITS	Intelligent Transportation System
LiFi	Light Fidelity
LoRa	Long Range
LoRaWAN	Long Range Wide Area Network
LPWAN	Low Power Wide Area Network
PWM	Pulse Width Modulation
RSSI	Received Signal Strength Indicator
RU	Road Unit
SDG	Sustainable Development Goal
SNR	Signal-to-Noise Ratio
V2I	Vehicle-to-Infrastructure
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything
V-IoT	Vehicular Internet of Things
VU	Vehicle Unit

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The rapid growth of vehicular traffic has led to a significant increase in road accidents, particularly in sensitive zones such as school areas, hospitals, residential streets, and construction sites ^[8]. Conventional speed control mechanisms — including static signboards and speed breakers — are often ineffective, inconvenient, and unable to adapt to dynamic traffic conditions. With advancements in the Internet of Things and wireless communication technologies, it has become possible to design intelligent systems that automatically regulate vehicle speed in a safer and more efficient manner ^[4].

This project proposes an IoT-based Intelligent Vehicle Speed Limiting and Monitoring System that leverages a hybrid communication approach combining Infrared (IR) sensing, Light Fidelity (LiFi) optical communication, and Long Range (LoRa) wireless telemetry. The roadside unit detects approaching vehicles via an IR sensor and transmits the permissible speed limit through a modulated LiFi signal. The vehicle's onboard unit receives and decodes this signal and enforces the limit through PWM-based motor speed control. LoRa is used in parallel for continuous transmission of vehicle speed and status to a remote monitoring station ^[3].

1.2 INTERNET OF THINGS (IoT)

The Internet of Things (IoT) refers to a network of interconnected physical devices embedded with sensors, actuators, communication modules, and processing units that enable them to collect, exchange, and act upon data autonomously ^[11]. In modern transportation, IoT serves as a key technological enabler for developing intelligent, adaptive, and connected vehicular systems ^[4]. By integrating vehicles, roadside infrastructure, and control platforms through IoT, transportation systems can evolve beyond passive monitoring toward real-time decision-making and automated control.

IoT systems operate by continuously sensing physical parameters — such as speed, proximity, and environmental conditions — and transmitting the sensed data through wireless communication channels to processing units or cloud platforms ^[11]. In vehicular applications, this enables vehicles to respond dynamically to road conditions, traffic regulations, and safety constraints without relying solely on driver intervention ^[7].

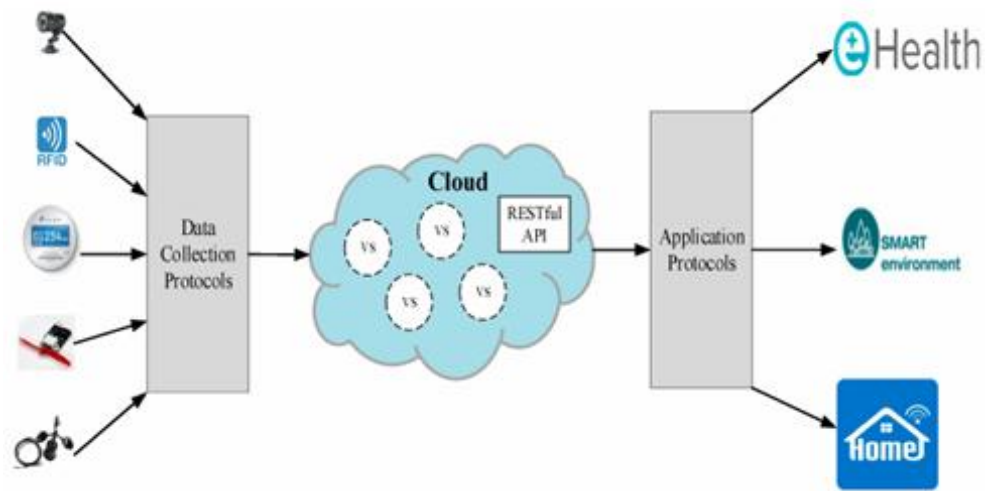


Figure 1.1: Cloud-Based IoT Framework

1.2.1 Types of IoT

- Consumer IoT: Used in everyday applications such as smart home devices, wearables, and personal monitoring systems.
- Commercial IoT: Applied in transportation, healthcare, and logistics, including vehicle monitoring and smart traffic management.
- Industrial IoT (IIoT): Used in industrial automation, smart agriculture, and energy management.
- Infrastructure IoT: Focuses on smart cities, intelligent transportation systems, and urban traffic control.
- Vehicular IoT (V-IoT): A specialised IoT domain integrating vehicles with communication networks for intelligent vehicle control and safety.

1.2.2 Applications of IoT

- Intelligent Transportation Systems: Real-time vehicle-infrastructure communication for traffic monitoring and adaptive speed control.
- Smart Speed Regulation: Automatic enforcement of speed limits in restricted zones via wireless control signals.
- Fleet Monitoring and Management: IoT-equipped vehicles tracked and managed for safety and operational efficiency.
- Smart Cities: Vehicles integrated with traffic lights, roadside sensors, and control centres to improve traffic flow.
- Connected and Autonomous Vehicles: IoT forms the backbone for data exchange required in advanced driver assistance systems.

1.3 LiFi TECHNOLOGY

Light Fidelity (LiFi) is a wireless communication technology that uses visible light or infrared radiation to transmit data, in contrast to conventional radio-frequency-based communication ^[12]. LiFi operates by modulating the intensity of a light source — typically a high-brightness LED — at speeds imperceptible to the human eye ^[19]. A photodetector on the receiving end captures these intensity variations and converts them back into electrical signals for processing.

LiFi offers several advantages for vehicular communication, including immunity to radio-frequency interference, inherent directionality that reduces interception risks, and high-speed data transfer at very low latency ^[14]. In this project, LiFi serves as the primary control channel for transmitting speed limit data from the roadside unit to the vehicle onboard unit. The directional nature of LiFi ensures that only vehicles within the coverage zone receive the speed limit signal, making it highly relevant for zone-based speed enforcement ^[15].

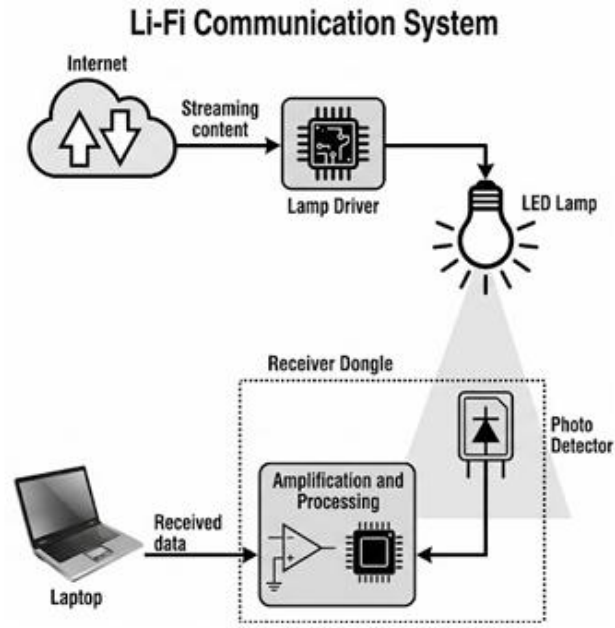


Figure 1.2: High-Level Architecture of a Li-Fi Communication System

1.3.1 Key Characteristics of LiFi

- High-speed optical data transmission using modulated LED signals.
- Inherently directional — reduces unintended signal reception.
- Immune to electromagnetic interference from radio frequency sources.
- Low-cost implementation using standard high-brightness LEDs and photodiodes.
- Suitable for short-range, point-to-area communication in constrained zones.

1.4 LoRa TECHNOLOGY

LoRa (Long Range Radio) is a wireless communication technology designed for long-range, low-power data transmission over unlicensed radio frequency bands ^[9]. Based on Semtech's Chirp Spread Spectrum (CSS) modulation technique, LoRa achieves remarkable range — several kilometres in open environments — while consuming minimal power ^[21]. It operates at frequency bands such as 433 MHz, 868 MHz, and 915 MHz depending on regional regulations ^[21].

LoRa's key advantage lies in its ability to maintain communication with moving vehicles over extended distances, even in environments with signal attenuation and multipath propagation ^[6]. Experimental studies have confirmed that LoRa communication performance is largely unaffected by vehicle speed or Doppler effects, validating its suitability for vehicular IoT applications ^[3]. In this project, LoRa is used exclusively for telemetry — continuously transmitting the vehicle's current speed and system status to a remote monitoring station.

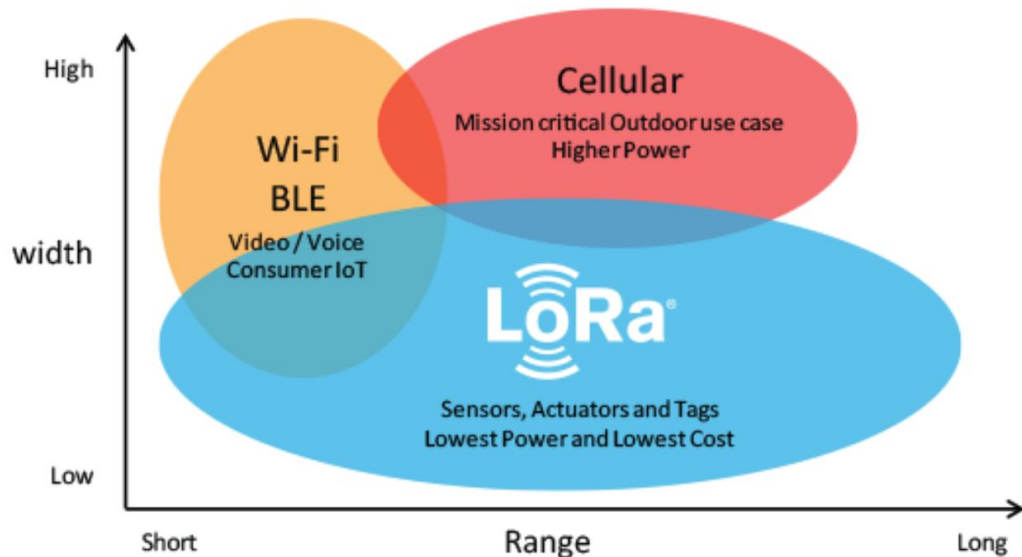


Figure 1.3: Wireless Communication Technologies based on Bandwidth and Transmission Range

1.4.1 Key Characteristics of LoRa

- Long-range communication: kilometres of range with minimal infrastructure.
- Low power consumption: ideal for battery-operated vehicular modules.
- Robustness to Doppler shift and signal fading in mobile environments.
- Configurable spreading factors for balancing range and data rate.
- Operates on unlicensed ISM bands — no subscription or SIM required.

1.5 INFRARED COMMUNICATION

Infrared (IR) communication involves the transmission and detection of electromagnetic radiation in the infrared spectrum, typically at wavelengths between 700 nm and 1 mm ^[13]. IR sensors are widely used in proximity detection, object detection, and short-range communication applications ^[17]. In this project, an IR sensor module is deployed at the roadside unit to detect the presence of an approaching vehicle within the detection zone.

When a vehicle enters the IR sensor's detection range, the sensor output triggers the microcontroller to activate LiFi transmission. This ensures that speed limit data is only broadcast when a vehicle is actually present, conserving energy and preventing spurious signals. The use of IR for detection is highly cost-effective, reliable, and requires minimal processing overhead, making it ideal as the first layer of the hybrid communication architecture ^[13].

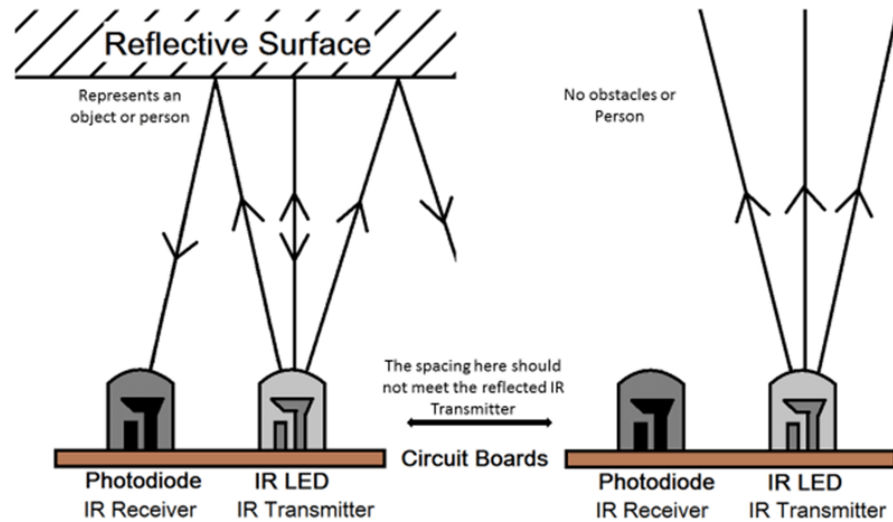


Figure 1.4: Working Principle of Infrared (IR) Sensor Module

1.6 VEHICULAR SYSTEMS

Vehicular systems refer to the integrated technologies, components, and control mechanisms used in modern vehicles to enhance safety, efficiency, comfort, and intelligence ^[20]. At the core of vehicular systems lies the concept of vehicle automation and connectivity, where vehicles interact not only with internal subsystems but also with external infrastructure and other vehicles. Modern vehicular systems combine hardware such as sensors, actuators, motor controllers, and communication modules with sophisticated embedded software to achieve intelligent operation.

The emergence of Intelligent Transportation Systems (ITS) has expanded the scope of vehicular systems by integrating vehicles into a broader transportation ecosystem ^[7]. In ITS,

vehicles communicate with roadside units, traffic management centres, and monitoring platforms to exchange information related to traffic conditions, speed limits, and road hazards [4]. This connectivity allows vehicles to adapt dynamically to changing road environments, reducing accidents and congestion.

1.6.1 Types of Vehicular Systems

- Safety Systems: Anti-lock Braking Systems (ABS), airbag control, electronic stability control, and collision avoidance.
- Control and Powertrain Systems: Engine control units, motor controllers, and electronic throttle control.
- Driver Assistance Systems (ADAS): Adaptive cruise control, lane departure warning, and automated parking.
- Vehicular Communication Systems: V2V, V2I, and V2X communication for real-time information exchange.

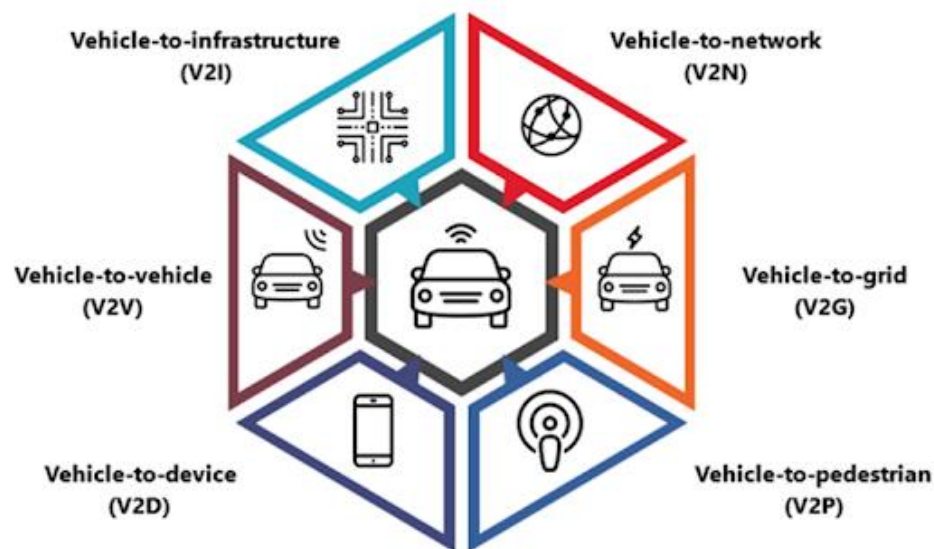


Figure 1.5: Vehicle-to-Everything (V2X) Communication Domains

- Infotainment and Comfort Systems: Navigation, entertainment, climate control, and human-machine interfaces.

1.7 OBJECTIVE OF THE PROJECT WORK

This project presents an Intelligent Vehicle Speed Limiting and Monitoring System that leverages a hybrid communication approach combining Infrared (IR), Light Fidelity (LiFi), and Long Range (LoRa) technologies. The system is designed to automatically regulate the speed of a robotic vehicle upon entering predefined restricted zones, thereby enhancing road safety and reducing dependence on driver intervention. A roadside unit detects vehicle presence via an IR

sensor and transmits the permissible speed limit using LiFi-based optical communication, while the vehicle unit receives, processes, and enforces the limit through PWM-based motor control.

The specific objectives of the project are:

- To design and implement a hybrid IR-LiFi-LoRa communication-based system that automatically enforces vehicle speed limits in restricted zones without driver intervention.
- To utilise LiFi-based optical communication for real-time, low-cost speed limit transmission from roadside units to vehicles.
- To implement PWM-based motor speed control for smooth and automatic speed regulation upon entering restricted zones.
- To enable remote monitoring of vehicle speed and system status via LoRa communication to a monitoring station.
- To provide a fail-safe mechanism and manual override feature ensuring safety under signal loss or emergency conditions.

1.8 NEED OF THE PROJECT WORK

The need for this system arises from the persistent challenge of over-speeding in sensitive zones, which continues to cause preventable accidents and fatalities. Existing speed control methods are either passive (signboards), physically disruptive (speed breakers), reactive (surveillance cameras), or prohibitively expensive for widespread deployment (V2X, 5G). None of these approaches autonomously and proactively enforce speed limits in real time at the vehicle level.

- **Lack of Real-Time Enforcement:** Most electronic systems detect violations after they occur and rely on post-event penalties. This reactive approach does not prevent accidents at the moment of risk. A proactive, real-time intervention mechanism is essential for zones where split-second speed compliance can save lives.
- **High Cost of Existing Solutions:** Advanced vehicle-to-everything (V2X) and GPS-based systems require dedicated roadside infrastructure, continuous network subscriptions, and complex onboard hardware. Their high cost makes them impractical for deployment at every school, hospital, or construction zone, particularly in developing regions.
- **Driver Dependence and Human Error:** Systems that rely on driver warnings or alerts are inherently limited by human reaction time, attentiveness, and intention. Automated speed enforcement that removes driver dependence is significantly more reliable in preventing over-speeding incidents.

- **Scalability Gap:** There is a clear requirement for a cost-effective, modular, and scalable solution that can be deployed across diverse road environments. The proposed hybrid IR-LiFi-LoRa system addresses this gap by using low-cost, widely available components and localized communication that does not depend on centralised network infrastructure.

1.9 SCOPE OF THE PROJECT

The proposed project focuses on developing a hybrid communication-based intelligent speed control system that combines real-time vehicle detection, optical data transmission, embedded motor control, and long-range telemetry into a unified and cost-effective prototype. At its core, the system emphasizes autonomous speed enforcement in restricted zones, where the vehicle's speed is automatically regulated based on the speed limit received from the roadside unit — without any dependence on driver intervention or continuous network infrastructure.

Key aspects of this module include:

- Real-time detection of vehicles entering restricted zones using IR sensing
- Zone-specific speed limit transmission via LiFi-based optical communication
- Automatic motor speed adjustment through PWM control upon signal reception

In addition to basic speed control, the system integrates a fail-safe and override decision mechanism to handle edge cases in real-world operation. Rather than acting as a passive speed limiter, the system actively monitors signal continuity, driver inputs, and system status to make control decisions in real time.

This decision and control mechanism enables:

- Automatic fallback to a safe default speed upon LiFi signal loss
- Timed manual override allowing the driver to bypass speed limiting for 20 seconds in emergencies
- Seamless resumption of autonomous control after the override window expires

Another important focus of the project is remote monitoring and telemetry, which plays a critical role in enabling supervision of the vehicle without physical access. By continuously transmitting vehicle speed and system status via LoRa, the system ensures that a remote monitoring station remains informed of the vehicle's operational state at all times.

This is achieved through:

- Continuous LoRa-based transmission of vehicle speed and system status (normal / speed-limited / override)
- Real-time reception and logging at a remote monitoring station
- cloud IoT platform and mobile dashboard integration for centralised supervision

The system is implemented as a hardware prototype on an RC car chassis, which demonstrates the feasibility of the hybrid communication approach in a controlled and reproducible environment. This makes it convenient for academic evaluation, testing, and future scaling.

Advantages of this approach include:

- Low-cost implementation using widely available embedded hardware components
- Independence from cellular networks or dedicated communication infrastructure
- Modular architecture supporting future expansion to real vehicles and multi-zone deployments

At the same time, the project clearly defines its limitations to maintain focus and feasibility. It does not include:

- Integration with real vehicle throttle or braking systems, as such implementation requires automotive-grade hardware and safety validation
- Outdoor large-scale deployment testing, which would require weatherproofing, extended LoRa range calibration, and multi-RU coordination

Overall, the system is developed for academic and prototype demonstration purposes, showcasing how hybrid IR-LiFi-LoRa communication, embedded motor control, and IoT-based telemetry can be integrated into a single intelligent vehicular speed control platform.

Future enhancements may include:

- GPS-based zone verification and multi-zone speed transition
- CAN bus integration for deployment in real vehicles
- Machine learning-based adaptive speed limit adjustment based on traffic density and environmental conditions

1.10 ORGANIZATION OF THE CHAPTERS

The report is structured systematically to present the development and analysis of the proposed system in a clear and logical manner:

- **Chapter 1: Introduction** - We introduce the project concept and provide an overview of the key technologies employed — IoT, LiFi, LoRa, IR communication, and vehicular systems. The objectives and need for the project are also discussed in this chapter.
- **Chapter 2: Literature Review** - We discuss the detailed literature survey of existing systems and related works, examining their techniques, limitations, and the research gaps that motivate the proposed solution.
- **Chapter 3: System Requirements** - We list the hardware and software requirements of the project along with detailed descriptions of each component and tool used in the system.
- **Chapter 4: Existing System** - We analyse the existing vehicular speed regulation systems, discuss their architecture, and identify their key drawbacks and the resulting problem statement.
- **Chapter 5: Proposed System** - We present the proposed system — its overview, architecture diagram, workflow, and module descriptions — along with placeholders for the system diagrams.
- **Chapter 6: Implementation** - We describe the implementation of each module and present snapshots of the system in operation.
- **Chapter 7: Results and Discussion** - We present the results and discussion including performance analysis of the system under various test conditions.
- **Chapter 8: Conclusion and Future Work** - We conclude the project with a summary of contributions and outline directions for future enhancement.

CHAPTER 2

LITERATURE REVIEW

2.1 SURVEY OF THE RELATED WORKS

The rapid evolution of Intelligent Transportation Systems (ITS) has led to extensive research in vehicular communication, IoT integration, and automated speed control ^[4]. Several studies have explored wireless technologies for vehicle monitoring, speed regulation, and safety enhancement ^[7]. The literature surveyed in this chapter highlights existing approaches, their limitations, and the research gap that motivates the proposed hybrid IR-LiFi-LoRa system.

Early research in vehicular systems focused primarily on cellular-based technologies such as GSM, LTE, and 5G for vehicle tracking and control. While these technologies offer low latency and high reliability, their high operational and deployment costs limit their applicability in cost-sensitive scenarios ^[7]. Recent studies have shifted toward Low Power Wide Area Network (LPWAN) technologies such as LoRa and LoRaWAN for vehicular applications, which enable long-range communication with minimal power consumption ^[10].

2.1.1 LoRaWAN for Vehicular Networking: Field Tests for Vehicle-to-Roadside Communication

Authors: Gabriele Di Renzone, Stefano Parrino, Giacomo Peruzzi, Alessandro Pozzebon, Lorenzo Vangelista

Year: 2024

Description:

This paper investigates the feasibility of using LoRaWAN for vehicular networking, specifically vehicle-to-roadside (V2R) communication ^[3]. The authors conducted extensive field tests in two scenarios: a vehicle moving on an elliptical track and a vehicle travelling on a straight road at different speeds. Performance metrics such as packet loss and Received Signal Strength Indicator (RSSI) were analysed across various spreading factors ^[3]. The study demonstrated that LoRaWAN communication is largely unaffected by vehicle speed and Doppler effects, proving its robustness in mobile environments. The results confirm that LoRaWAN can reliably support vehicular IoT applications where low latency is not critical ^[3].

2.1.2 Secure Communication Frameworks for IoT-based Vehicular Systems, IEEE IoT Journal

Year: 2024

Description:

The research presents security challenges and solutions for IoT-enabled vehicles, including authentication, data encryption, and intrusion detection. It emphasizes the vulnerabilities of RFID-based solutions and recommends more secure, scalable protocols such as LoRaWAN. These insights justify the proposed shift from RFID to LoRa in our project, enhancing both security and practicality in real-world deployment.

2.1.3 IoT-Based Smart Speed Control System for Vehicles

Authors: S. Vinodhini, S. Rajkumar, S. K. Subramaniam

Year: 2023

Description:

This paper proposes an IoT-based vehicle speed control system aimed at reducing road accidents in sensitive areas ^[4]. The system uses sensors and wireless communication to detect restricted zones and automatically control vehicle speed. The authors emphasise automation in speed regulation to minimise human error ^[4]. Experimental results indicate improved compliance with speed limits in controlled environments. However, the proposed system relies on short-range communication technologies and lacks scalability for large-area or high-speed vehicular scenarios.

2.1.4 An Experimental Evaluation of LoRa for Vehicular Communication

Authors: A. P. A. Torres, C. B. da Silva, H. Tertuliano Filho

Year: 2021

Description:

This research presents an experimental study on the use of LoRa technology for communication between moving vehicles and roadside infrastructure ^[20]. The authors evaluated packet delivery ratio and signal behaviour at different vehicle speeds and distances. The experiments showed that LoRa communication remains stable even when vehicles are in motion, especially when lower spreading factors are used ^[20]. The study highlights the suitability of LoRa for non-real-time vehicular applications such as monitoring and status reporting. However, the paper mainly focuses on communication performance and does not integrate any active vehicle control mechanism.

2.1.5 Vehicle Speed Monitoring and Control Using IoT and Embedded Systems

Author: R. Salazar-Cabrera, A. Pachon de la Cruz, J. M. Madrid Molina

Year: 2019

Description:

This work focuses on monitoring vehicle speed using IoT devices and embedded controllers^[16]. The system collects speed data and transmits it to a centralised monitoring platform for analysis and enforcement. Alerts are generated when vehicles exceed predefined speed limits^[16]. While the system improves speed monitoring and reporting, it does not directly enforce speed reduction at the vehicle level. Additionally, dependence on centralised infrastructure increases latency and operational cost.

2.1.6 Light Fidelity for Internet of Things: A Survey

Author: M. Petrosino, D. Striccoli, A. Romanov, G. Boggia, L. A. Grieco

Year: 2023

Description:

This paper presents a comprehensive survey of LiFi technology and its applicability in IoT environments. The authors examine the fundamental principles of LiFi-based optical wireless communication, including modulation techniques, channel characteristics, and system architectures suitable for IoT deployments. The survey analyses the advantages of LiFi over conventional radio-frequency communication, particularly its immunity to electromagnetic interference, inherent directionality, and high spectral efficiency. The paper also identifies key challenges in LiFi-IoT integration such as mobility management, ambient light interference, and line-of-sight requirements. However, the survey focuses primarily on indoor environments and does not address vehicular or outdoor deployment scenarios where LiFi could serve as a directional control channel.

2.1.7 Analysis of Li-Fi for Real-Time Vehicular Applications

Author: N. Singh, et al.

Year: 2023

Description:

This paper investigates the feasibility and performance of LiFi technology specifically for real-time vehicular communication applications. The authors analyse the optical channel characteristics between vehicle-mounted LED transmitters and photodetector receivers under dynamic traffic conditions. Key parameters including signal-to-noise ratio, data throughput, and reception reliability are evaluated at varying distances and angles of incidence. The study demonstrates that LiFi can achieve reliable short-range communication in vehicular environments, making it suitable for vehicle-to-infrastructure data exchange. However, the paper does not

incorporate any active speed control mechanism at the vehicle level and focuses exclusively on the communication layer performance without integration with embedded motor control systems.

2.1.8 OLO-IR-Free: An Improved Algorithm for Real-Time Detection of Vehicles in Infrared Images

Author: Z. Zhang, J. Huang, G. Hei, W. Wang

Year: 2023

Description:

This paper proposes an improved deep learning algorithm for real-time vehicle detection using infrared imaging. The authors evaluate the performance of infrared-based vehicle detection under varying environmental conditions including low visibility, nighttime, and adverse weather, demonstrating that infrared sensing provides significantly more reliable vehicle detection compared to visible-light camera systems in such scenarios. The study validates infrared sensing as a robust and low-cost mechanism for vehicle presence detection in intelligent transportation applications. However, the work focuses on image-based deep learning detection and does not explore simpler IR sensor-based proximity detection as used in embedded IoT systems, nor does it integrate any communication or speed control mechanism.

2.2 TECHNIQUES / ALGORITHMS

The reviewed works employ a diverse combination of IoT communication protocols, optical wireless communication techniques, infrared sensing methods, embedded control logic, and vehicular networking concepts — each contributing a distinct layer to the overall understanding of intelligent speed regulation systems.

LoRa and LoRaWAN Communication

LoRa and LoRaWAN are widely adopted in vehicular IoT research due to their long-range capability, low power consumption, and resilience to interference and Doppler effects ^{[3][6]}. These technologies are particularly suitable for vehicle-to-roadside communication in non-latency-critical applications such as telemetry and status reporting ^[5]. Field experiments have confirmed that LoRaWAN packet delivery remains stable across varying vehicle speeds and spreading factor configurations, validating its use as a reliable monitoring channel ^[3]. The configurable spreading factor and bandwidth parameters allow the system to balance range and data rate based on deployment requirements ^[10].

LiFi-Based Optical Communication

LiFi technology employs modulated visible light or infrared radiation for high-speed, directional, and interference-immune data transmission ^[12]. The core technique involves ON/OFF

Keying (OOK) or more advanced modulation schemes applied to LED drivers, where intensity variations are captured by photodetectors and decoded into digital data ^[12]. In vehicular contexts, LiFi has been evaluated for vehicle-to-infrastructure data exchange, with studies demonstrating reliable short-range communication at varying distances and angles of incidence ^[14]. Key parameters evaluated in LiFi vehicular research include signal-to-noise ratio, data throughput, and ambient light interference immunity ^[14]. The inherent directionality of the LiFi optical channel ensures zone-specific signal reception, making it particularly well suited for localized speed limit enforcement in restricted areas ^[13].

Infrared Sensing for Vehicle Detection

Infrared sensing is employed in intelligent transportation research both as a proximity detection mechanism and as an imaging modality for vehicle recognition ^[15]. Simple IR sensor modules — consisting of an emitter-receiver pair — detect vehicle presence through reflected or interrupted beam sensing, offering fast response times and minimal processing overhead ^[15]. Advanced infrared imaging techniques, such as the YOLO-IR-Free algorithm, extend the applicability of infrared sensing to real-time vehicle detection under low visibility and adverse weather conditions ^[13]. These studies confirm that infrared-based detection provides more robust and reliable vehicle presence identification compared to visible-light systems in challenging environments, supporting its adoption as the first detection layer in the proposed hybrid architecture.

Embedded Speed Control Logic

Speed control mechanisms in the reviewed works are generally implemented using embedded controllers interfaced with vehicle throttle systems or motor controllers. Rule-based control logic is commonly adopted, where predefined speed limits are enforced based on received signals or zone detection ^[4]. PWM-based motor speed regulation is a standard technique for smooth and proportional speed adjustment in embedded vehicular systems, enabling gradual deceleration without abrupt speed changes ^[4]. IoT-based speed control systems also incorporate alert generation and centralised reporting, though most lack direct vehicle-level enforcement capability ^[16].

Security Considerations in IoT Vehicular Systems

Security is an emerging concern in IoT-based vehicular communication. Research highlights vulnerabilities in RFID-based vehicle identification systems and recommends more secure, scalable protocols for zone-based access and speed control. The use of structured communication protocols — as employed in the proposed LiFi encoding scheme and LoRa telemetry format — provides a basis for future integration of lightweight authentication and encryption mechanisms to prevent signal spoofing and unauthorised speed limit injection.

Research Gap

Most existing approaches focus either on communication reliability or speed monitoring in isolation, but very few integrate detection, communication, and direct vehicle-level speed control into a single unified system ^[16]. No existing work was identified that combines IR-based vehicle detection, LiFi-based optical speed limit transmission, and LoRa-based long-range telemetry into a single hybrid architecture, confirming the novelty and contribution of the proposed system.

2.3 SUMMARY OF LITERATURE REVIEW

The literature review highlights that IoT-based vehicular systems have gained significant attention due to their potential to improve road safety and traffic efficiency ^[11]. Technologies such as LoRa and LoRaWAN have proven reliable for vehicle-to-roadside communication, especially in scenarios where data packets are small and latency requirements are moderate ^[2]. An important observation is that vehicle speed and mobility do not severely degrade LoRa communication performance when appropriate parameters are selected ^{[3][6]}.

However, most existing systems focus primarily on monitoring, data collection, or post-event analysis rather than proactive control ^[16]. Speed monitoring systems generally depend on alerts or centralised enforcement, requiring human intervention ^[4]. Systems that attempt automated speed control often rely on short-range communication or sensor-based detection, restricting their coverage ^[20]. No existing work was found that integrates IR-based detection, LiFi-based speed limit transmission, and LoRa-based telemetry into a single unified vehicular speed control system, confirming the novelty of the proposed approach.

Table 2.1: Survey of Literature Review

S.No.	Title of the Paper	Description	Issues / Drawbacks / Limitations
1	LoRaWAN for Vehicular Networking: Field Tests for V2R Communication [2024]	Evaluates LoRaWAN performance for vehicles in motion using field experiments at different speeds and spreading factors.	Focuses only on communication performance; no active vehicle speed control mechanism is implemented.

2	Secure Communication Frameworks for IoT-based Vehicular Systems [2024]	Examines security challenges in IoT-enabled vehicles and recommends scalable protocols such as LoRaWAN over RFID-based solutions.Focuses on security frameworks only; does not address real-time speed control or physical enforcement at the vehicle level.	Examines security challenges in IoT-enabled vehicles and recommends scalable protocols such as LoRaWAN over RFID-based solutions.Focuses on security frameworks only; does not address real-time speed control or physical enforcement at the vehicle level.
3	IoT-Based Smart Speed Control System for Vehicles [2023]	Proposes automatic speed regulation using IoT sensors and wireless communication to detect restricted zones.	Relies on short-range communication; limited scalability and coverage for real-world deployment.
4	An Experimental Evaluation of LoRa for Vehicular Communication [2021]	Studies LoRa reliability for moving vehicles at different speeds and distances, evaluating packet delivery ratio.	Does not implement any active vehicle control; focused only on communication layer evaluation.
5	Vehicle Speed Monitoring and Control Using IoT and Embedded Systems [2019]	Monitors vehicle speed and reports violations to a centralised platform using IoT devices and embedded controllers.	No direct speed enforcement at vehicle level; dependent on centralised infrastructure and human follow-up.
6	Light Fidelity for Internet of Things: A Survey [2023].	Surveys LiFi optical wireless communication principles, modulation techniques, and IoT system architectures.	Focuses primarily on indoor environments; does not address vehicular or outdoor LiFi deployment scenarios.
7	Analysis of Li-Fi for Real-Time Vehicular Applications [2023]	Evaluates LiFi channel characteristics for vehicle-to-infrastructure communication	No active speed control integration; focuses exclusively on

		including SNR, throughput, and reception reliability.	communication layer without embedded motor control.
8	YOLO-IR-Free: An Improved Algorithm for Real-Time Detection of Vehicles in Infrared Images [2023]	Proposes an infrared-based deep learning algorithm for real-time vehicle detection under low visibility and adverse weather.	Focuses on image-based deep learning detection; does not explore simple IR proximity sensing or speed control integration.

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 HARDWARE REQUIREMENTS

3.1.1 Minimum Requirements

- Microcontroller: Arduino Uno (roadside unit) / Arduino Uno or ESP32 (vehicle unit)
- IR Sensor Module (basic obstacle detection)
- Standard LED (LiFi transmitter)
- Photodiode (LiFi receiver)
- L298N Motor Driver
- DC Motors with basic RC car chassis
- LoRa Module SX1278 (433 MHz)
- USB Cable for programming
- 9V / 12V Battery Pack

3.1.2 Recommended Requirements

- Microcontroller: ESP32 (dual-core, hardware PWM support) for both units
- High-brightness LED with current limiting resistor (LiFi transmitter)
- Phototransistor with amplifier circuit (improved LiFi signal sensitivity)
- TB6612FNG Motor Driver (lower power dissipation, higher efficiency)
- DC Motors with sturdy RC car chassis (adequate torque and current rating)
- LoRa Module SX1278 with antenna (extended telemetry range)
- Buzzer / LED Alert Indicators
- I2C Display for Alert and Status Indication
- Dedicated Battery Pack with adequate current rating per unit
- Breadboard / Custom PCB with jumper wires for stable connections

3.2 SOFTWARE REQUIREMENTS

3.2.1 Minimum Requirements

- Operating System: Windows 10 (64-bit)
- IDE: Arduino IDE (v1.8 or above)
- Programming Language: Embedded C / Arduino Framework
- LoRa Library (SX1278 compatible)

- Basic Serial Monitor (for debugging)

3.2.2 Recommended Requirements

- Operating System: Windows 10 / 11 (64-bit)
- IDE: Visual Studio Code with PlatformIO Extension
- Programming Language: Embedded C / Arduino Framework
- Libraries: LoRa Library (SX1278), ESP32 LEDC PWM Functions
- Cloud IoT Platform (for telemetry logging and remote monitoring)
- Mobile Dashboard Application (for vehicle control and data visualisation)

3.3 ABOUT THE HARDWARE

3.3.1 ESP32 Microcontroller

The ESP32 is a powerful, low-cost System-on-Chip (SoC) microcontroller developed by Espressif Systems, featuring a dual-core Xtensa LX6 processor running at up to 240 MHz ^[11]. It integrates Wi-Fi and Bluetooth connectivity alongside a rich set of peripherals including multiple UART, SPI, I2C, and PWM interfaces. The ESP32 is compatible with the extensive Arduino library ecosystem and highly suitable for rapid IoT prototyping ^[11].

In this project, the ESP32 serves as the primary microcontroller for the vehicle onboard unit. It is responsible for receiving and decoding the LiFi signal via the photodiode interface, computing PWM duty cycle adjustments for motor speed control, managing the manual override logic, and transmitting telemetry data via the LoRa module. Its hardware PWM (LEDC) peripheral enables smooth and precise motor speed regulation, which is critical for gradual and safe speed reduction.

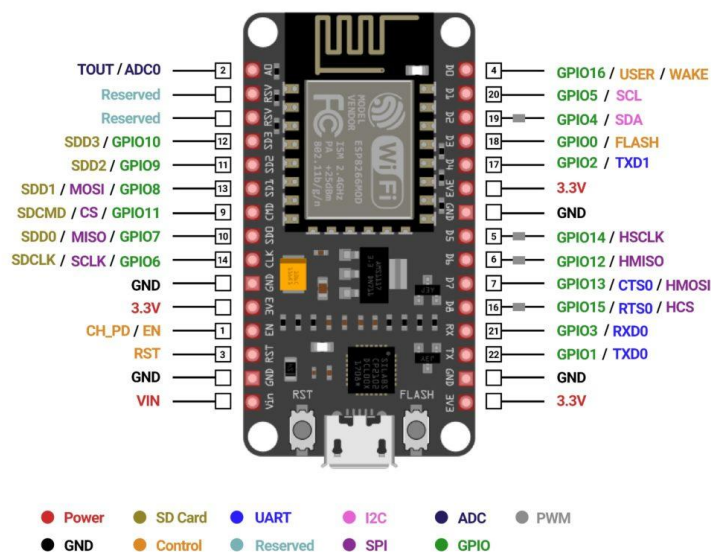


Figure 3.1: ESP32 Microcontroller

3.3.2 IR Sensor Module

The IR sensor module consists of an infrared LED emitter and a photodetector receiver pair. When an object passes between or in front of the sensor, the reflected or interrupted infrared beam changes the sensor output state ^[13]. These sensors are widely used in proximity and object detection applications owing to their low cost, fast response time, and simple interface ^[17].

In the proposed roadside unit, the IR sensor continuously monitors the road ahead. When a vehicle enters the detection zone, the sensor output triggers the microcontroller, which then activates the LiFi LED transmitter to begin broadcasting the speed limit signal. This ensures the LiFi transmission is only active when a vehicle is actually detected, conserving energy and reducing false signals.

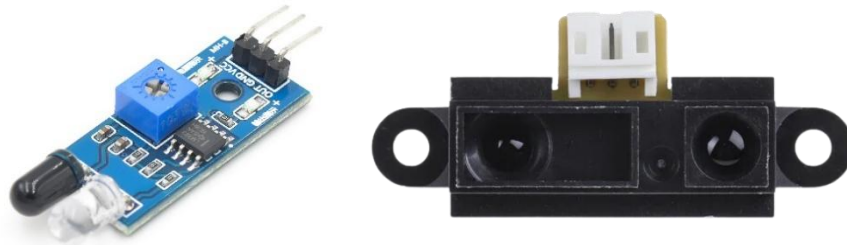


Figure 3.2: IR Sensor Module

3.3.3 High-Brightness LED (LiFi Transmitter)

A high-brightness LED serves as the LiFi transmitter in the roadside unit. The microcontroller modulates the LED's ON/OFF state at high frequencies imperceptible to the human eye to encode digital data representing the speed limit ^[12]. Binary encoding is used, where specific pulse patterns represent start bits, data bits, and stop bits — forming a simple but effective optical communication protocol ^[15].

LEDs are ideal for LiFi transmission due to their fast-switching speeds (in the microsecond range), low power consumption, long operational lifespan, and extremely low cost ^[19]. The directional nature of the LED's emission cone ensures that the speed limit signal is only receivable by vehicles within the designated zone, making this approach particularly relevant for point-to-area speed enforcement ^[14].



Figure 3.3: High-Brightness LED (LiFi Transmitter)

3.3.4 Photodiode / Phototransistor (LiFi Receiver)

A photodiode or phototransistor serves as the LiFi receiver on the vehicle onboard unit. These components convert incident light intensity into a proportional electrical signal, enabling the microcontroller to detect and decode the modulated LiFi signal from the roadside LED ^[12]. A phototransistor is preferred for its higher sensitivity and current output compared to a standard photodiode, making signal decoding more reliable in ambient light conditions ^[18].

An optional amplifier circuit may be added to boost the signal amplitude and improve noise immunity, particularly in outdoor environments where ambient light may interfere with reception ^[14]. The decoded signal is then processed to extract the transmitted speed limit value.

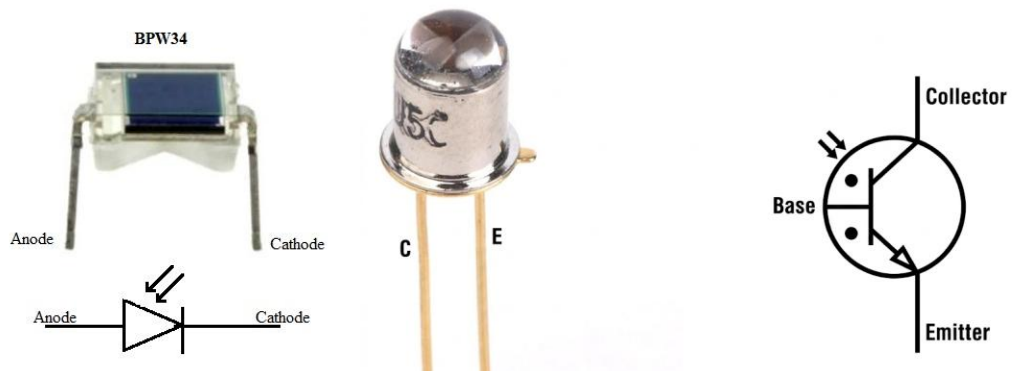


Figure 3.4: Photodiode / Phototransistor (LiFi Receiver)

3.3.5 L298N Motor Driver

The L298N is a dual H-bridge motor driver module commonly used for controlling DC motors and stepper motors. It can drive two DC motors independently with a maximum current of up to 2 A per channel (with proper heat dissipation). Unlike more modern drivers, the L298N uses bipolar junction transistors (BJTs), which results in higher power losses and lower efficiency compared to MOSFET-based drivers, but it remains widely used due to its robustness and ease of implementation ^[4].

In the proposed vehicle unit, the L298N interfaces between the ESP32 and the DC drive motors of the RC car chassis. PWM signals from the ESP32's LEDC peripheral are fed to the motor driver's input pins to control motor speed. When the speed limit signal is received and decoded, the ESP32 reduces the PWM duty cycle, causing the motor driver to proportionally reduce motor speed, achieving smooth and controlled speed regulation.

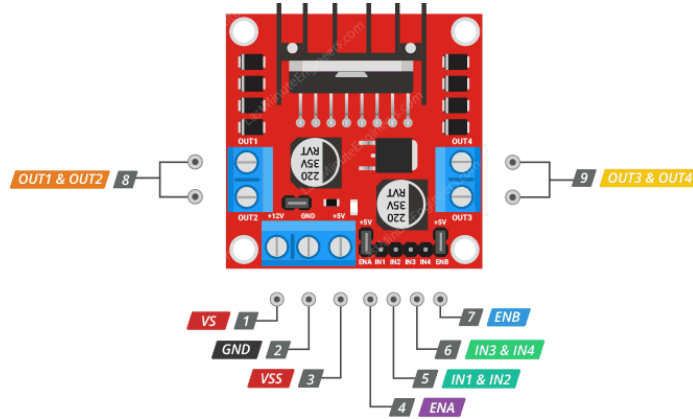


Figure 3.5: L298N Motor Driver

3.3.6 LoRa Module (SX1278)

The SX1278 is a LoRa transceiver module operating at 433 MHz, based on Semtech's LoRa Chirp Spread Spectrum modulation technology ^[21]. It supports a communication range of several kilometres in open environments and offers configurable spreading factors, bandwidth, and coding rates to balance range and data throughput ^[2]. The module communicates with the host microcontroller via SPI interface.

In this project, the LoRa module is mounted on the vehicle onboard unit and transmits vehicle speed and system status (normal / speed-limited / override active) to a remote monitoring station equipped with a corresponding LoRa receiver ^[3]. This telemetry capability enables remote supervision of the vehicle without requiring cellular network connectivity, making the system self-contained and infrastructure-independent ^[5].

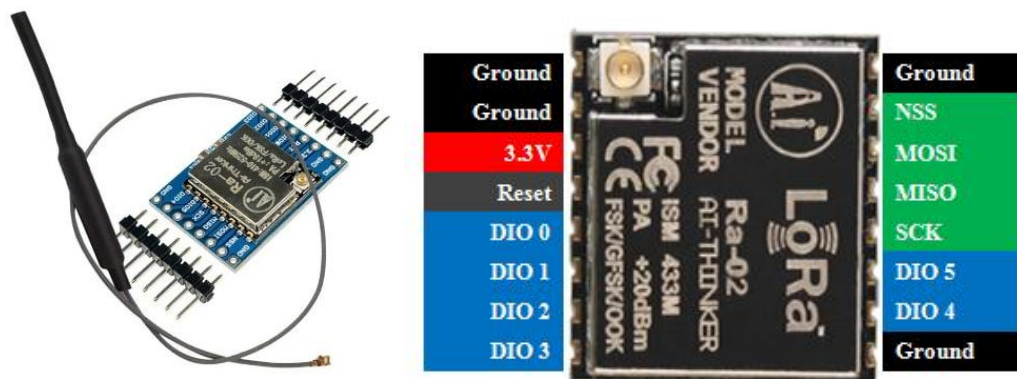


Figure 3.6: LoRa SX1278 Module

3.3.7 16x2 I2C LCD Display

The 16x2 I2C LCD Display is a commonly used output device in embedded systems and IoT applications for displaying real-time information ^[11]. It is capable of showing 16 characters per line across 2 lines, making it suitable for displaying sensor values, system status, alerts, and user messages efficiently.

In this project, the 16x2 I2C LCD Display is used to provide visual feedback from the vehicular monitoring system. The I2C communication interface reduces the number of required microcontroller pins, simplifying circuit connections and improving system integration. The display is used to show important parameters such as sensor readings, connectivity status, and operational messages during system testing and monitoring.

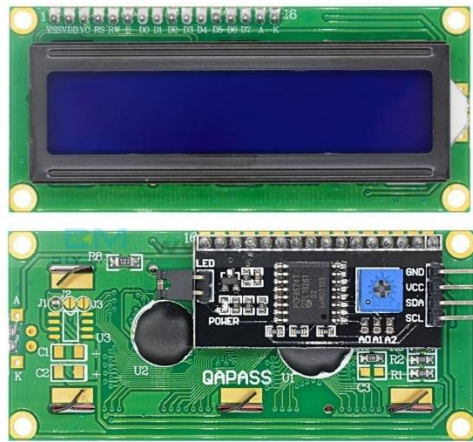


Figure 3.7: 16x2 I2C LCD Display

3.3.8 Intel Core i5 Processor

Intel Core i5 processors are widely used in desktop and laptop computers, offering a balance between performance and power efficiency. These processors are capable of handling development tasks such as code compilation, simulation, and system monitoring efficiently.

A system equipped with an Intel Core i5 processor is used as the development and testing platform. It supports running development environments such as Arduino IDE and Visual Studio Code, as well as cloud interfaces for monitoring and debugging the IoT-based vehicular system.

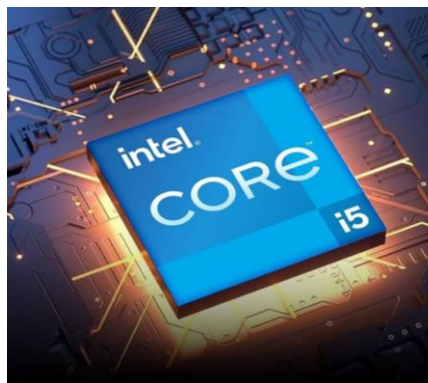


Figure 3.8: Intel i5

3.4 ABOUT THE SOFTWARE

3.4.1 Visual Studio Code with PlatformIO

Visual Studio Code (VS Code) is a lightweight, powerful, and extensible source code editor developed by Microsoft. PlatformIO is an open-source embedded development ecosystem that integrates seamlessly with VS Code, providing a unified development environment for embedded systems programming across a wide range of microcontroller platforms including ESP32 and Arduino ^[11].

In this project, VS Code with PlatformIO serves as the primary development environment for writing, compiling, uploading, and debugging the embedded firmware for both the roadside unit and the vehicle onboard unit. PlatformIO's library manager simplifies the integration of the LoRa SX1278 library and ESP32-specific LEDC PWM functions. The integrated serial monitor enables real-time debugging and observation of communication events, motor speed values, and system status messages during testing.

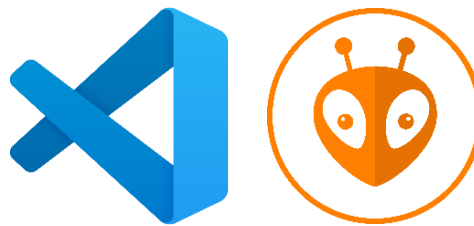


Figure 3.9: Visual Studio Code and Platform IO Extension

3.4.2 Embedded C / Arduino Framework

The Arduino framework provides a simplified, hardware-abstraction layer over the underlying ESP32 hardware, enabling developers to write embedded firmware using familiar C/C++ syntax with a rich set of pre-built libraries. The framework manages peripheral initialization, interrupt handling, and communication interfaces, allowing the development focus to remain on application logic ^[11].

For this project, the Arduino framework is used to implement the LiFi signal decoding algorithm, the PWM motor speed control logic, the LoRa telemetry transmission routine, and the manual override timer mechanism. The structured and modular nature of Arduino-compatible code facilitates easier debugging, testing, and future extension of the system.

3.4.3 Cloud IoT Platform

A cloud IoT platform is incorporated as a component for centralised monitoring and configuration of the system. The platform receives telemetry data from the LoRa monitoring station, stores speed and status logs, and provides a web or mobile-accessible dashboard for remote

supervision ^[11]. This enables administrators to monitor multiple vehicles and roadside units from a single interface without physical access to the hardware.

3.4.4 Mobile Dashboard

A mobile dashboard application complements the cloud platform by providing a user-friendly interface for controlling the RC car setup and visualising real-time telemetry data ^[4]. The specific platform and dashboard tools are to be finalised as the project progresses, with candidate options including established IoT cloud platforms and mobile visualisation tools.

CHAPTER 4

EXISTING SYSTEM

4.1 EXISTING SYSTEM OVERVIEW

The existing system of vehicular speed regulation encompasses a range of approaches — from purely physical mechanisms to electronically assisted monitoring systems — each with inherent limitations in terms of adaptability, automation, and real-time enforcement ^[8]. Understanding these limitations is essential for motivating the design of the proposed hybrid system.

Conventional Speed Control Methods

- Static speed limit signboards are installed in sensitive zones to inform drivers of permissible speeds, but their effectiveness depends entirely on driver awareness and compliance ^[8].
- Speed breakers are widely used to physically force vehicles to slow down, but they are permanent structures that cannot adapt to time-based or situational requirements and cause discomfort to all vehicles regardless of their actual speed.
- Manual enforcement by traffic authorities depends on human presence and is not scalable for continuous, round-the-clock coverage across all sensitive zones ^[4].

Electronic Speed Monitoring Systems

- Radar-based speed detection and surveillance cameras are used to identify over-speeding vehicles and record violations ^[4].
- These systems impose penalties after the incident rather than preventing unsafe speed in real time, limiting their effectiveness as accident prevention tools ^[8].
- High installation and maintenance costs restrict their deployment to major roads and intersections only.

GPS-Based Speed Assistance Systems

- Modern vehicles are equipped with GPS-enabled navigation systems that identify speed limits based on digital map data and alert drivers when limits are exceeded ^[8].
- Their performance depends on accurate and up-to-date map data, continuous satellite signal availability, and timely driver response.
- These systems are advisory in nature and do not enforce speed reduction automatically ^[8].

V2X and Network-Based Vehicular Systems

- Vehicle-to-Everything (V2X) communication models enable data exchange between vehicles, infrastructure, and traffic management systems using cellular networks, DSRC, or 5G [7].
- These systems are capable of real-time speed information delivery but require expensive dedicated roadside infrastructure, continuous network coverage, and complex onboard hardware [7].
- Their high cost and infrastructure dependency make them impractical for widespread deployment, particularly in developing regions or temporary restricted zones.

4.2 EXISTING SYSTEM ARCHITECTURE

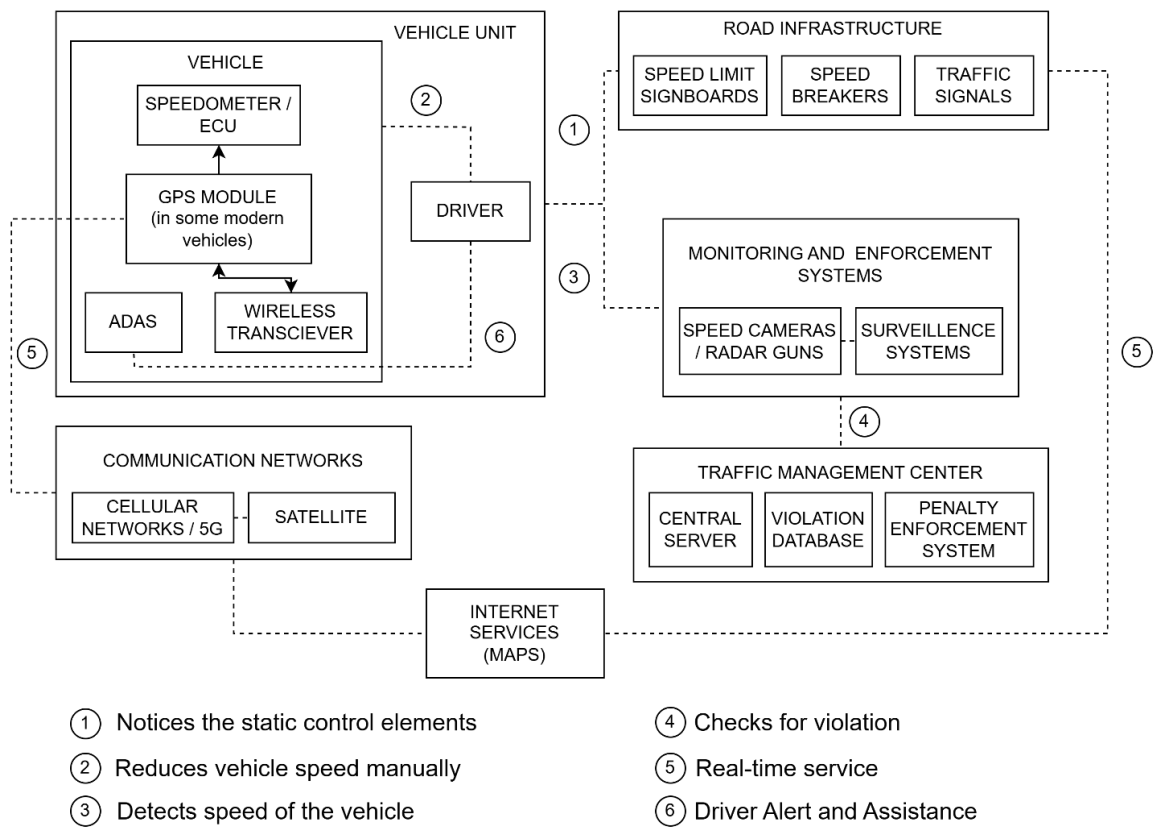


Figure 4.1: Existing System Diagram

Figure 4.1 represents the existing framework associated with vehicular speed regulation. The existing vehicular speed regulation architecture is a combination of physical infrastructure, onboard vehicle systems, and centralized monitoring platforms [8]. At the vehicle level, speed is primarily controlled by the driver, with basic assistance provided by speedometers and, in modern vehicles, GPS-based speed advisory systems. These systems can warn the driver when speed limits are exceeded but do not enforce speed reduction automatically [8].

Roadside infrastructure such as speed limit signboards and speed breakers act as static control elements. While signboards rely on driver compliance, speed breakers physically force

vehicles to slow down. Both methods lack adaptability and cannot respond to real-time traffic conditions or temporary restrictions [4].

Electronic monitoring systems such as speed cameras and radar guns are placed along roads to detect violations. These systems transmit data through communication networks to centralized traffic management centers [4]. The collected data is stored in databases and used for post-event enforcement such as issuing fines or penalties.

Advanced architectures incorporate GPS satellites and V2X communication networks to provide real-time information exchange between vehicles and infrastructure [7]. These systems depend on continuous cellular or dedicated communication networks, making them expensive to deploy and maintain [5]. Additionally, system functionality is often compromised in areas with poor network coverage [6].

Overall, the existing system architecture is **monitoring-centric and enforcement-driven**, with limited real-time control over vehicle speed [8]. The heavy reliance on infrastructure and continuous connectivity highlights the need for a simpler, cost-effective, and autonomous speed control mechanism.

4.3 PROBLEM STATEMENT

Road traffic accidents caused by over-speeding remain a critical safety concern, particularly in sensitive zones such as school areas, hospitals, residential streets, and construction sites [8]. Existing speed regulation mechanisms are either static, reactive, or dependent on expensive and continuous network infrastructure. They lack the ability to autonomously enforce speed limits in real time without driver intervention [4].

- **Ineffective speed regulation:** Conventional methods such as signboards and speed breakers do not guarantee compliance and cannot dynamically enforce speed limits based on changing conditions [8].
- **Lack of real-time enforcement:** Most electronic systems focus on post-violation penalties rather than proactive prevention, limiting their effectiveness in reducing accidents at the moment they occur [4].
- **Limited automation:** Current systems provide warnings or monitoring but lack automated mechanisms to directly control vehicle speed without driver intervention [16].
- **High dependency on network infrastructure:** Advanced GPS and V2X-based solutions require continuous high-bandwidth connectivity, making them costly and unreliable in areas with poor network coverage [7].

- **Scalability and deployment challenges:** The cost and complexity of existing intelligent transportation systems restrict their adoption across different regions and road types ^[5].
- **Need for a cost-effective solution:** There is therefore a clear requirement for an economical, scalable, and autonomous system capable of dynamically controlling vehicle speed in real time, which the proposed hybrid IR-LiFi-LoRa system directly addresses ^[3].

CHAPTER 5

PROPOSED SYSTEM

5.1 PROPOSED SYSTEM OVERVIEW

The proposed system presents an IoT-based dynamic vehicle speed control mechanism designed to automatically regulate vehicle speed in restricted and sensitive zones. Unlike conventional systems that rely on driver compliance or post-event enforcement, the proposed solution actively enforces speed limits in real time through a hybrid communication approach combining Infrared (IR) detection, LiFi optical communication, and LoRa wireless telemetry [8].

Road control units are deployed in predefined restricted zones such as school areas, hospitals, accident-prone locations, and construction zones [4]. Each roadside unit continuously monitors for approaching vehicles using an IR sensor.

Upon detecting a vehicle, the microcontroller activates a high-brightness LED to transmit the permissible speed limit using LiFi-based modulated optical communication [12]. The vehicle onboard unit — mounted on a robotic RC car — receives this optical signal through a photodiode, decodes the speed limit value, and automatically adjusts the motor speed via PWM control to comply with the received limit [15].

Simultaneously, the vehicle onboard unit continuously transmits its current speed and system status — normal operation, speed-limited, or override active — to a remote monitoring station using the LoRa module [3]. This enables real-time supervision without requiring cellular network connectivity.

A manual override button is provided on the vehicle unit, allowing the driver to temporarily bypass the speed restriction for 10 seconds in emergency situations, after which the system automatically resumes autonomous control. A fail-safe mechanism ensures that if the LiFi signal is lost, the vehicle maintains the last received speed limit or switches to a predefined safe default speed.

5.2 PROPOSED SYSTEM ARCHITECTURE

Figures 5.1 and 5.2 represent the proposed framework associated with vehicular speed regulation. The proposed system architecture consists of three principal components: the Road Unit (RU), the Vehicle Unit (VU), and a Cloud-based Monitoring Platform [11]. The RU is installed at the boundary of each restricted speed zone and handles vehicle detection and speed limit transmission. The VU is mounted on the RC car and handles signal reception, speed control, telemetry, and override management. The cloud platform provides centralised monitoring and zone configuration.

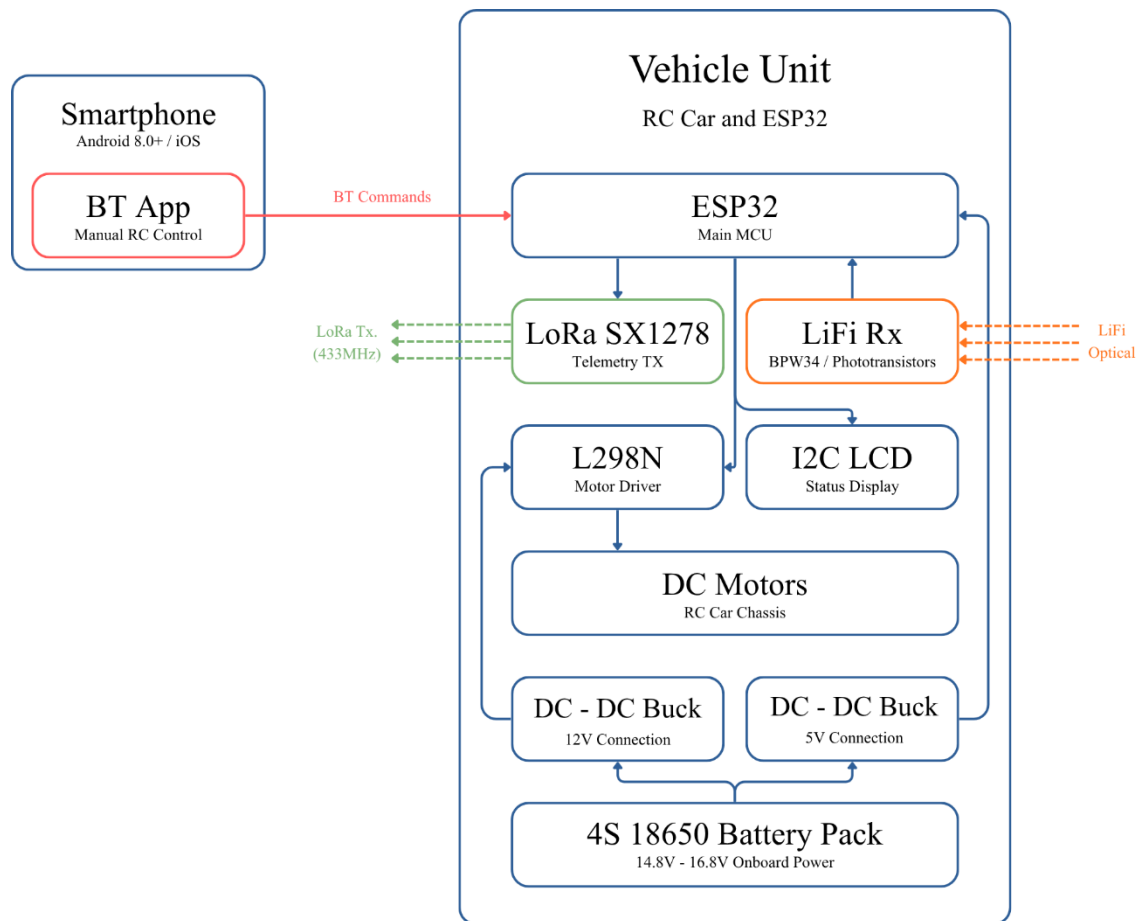


Figure 5.1: Proposed System Architecture Diagram – Vehicle Unit

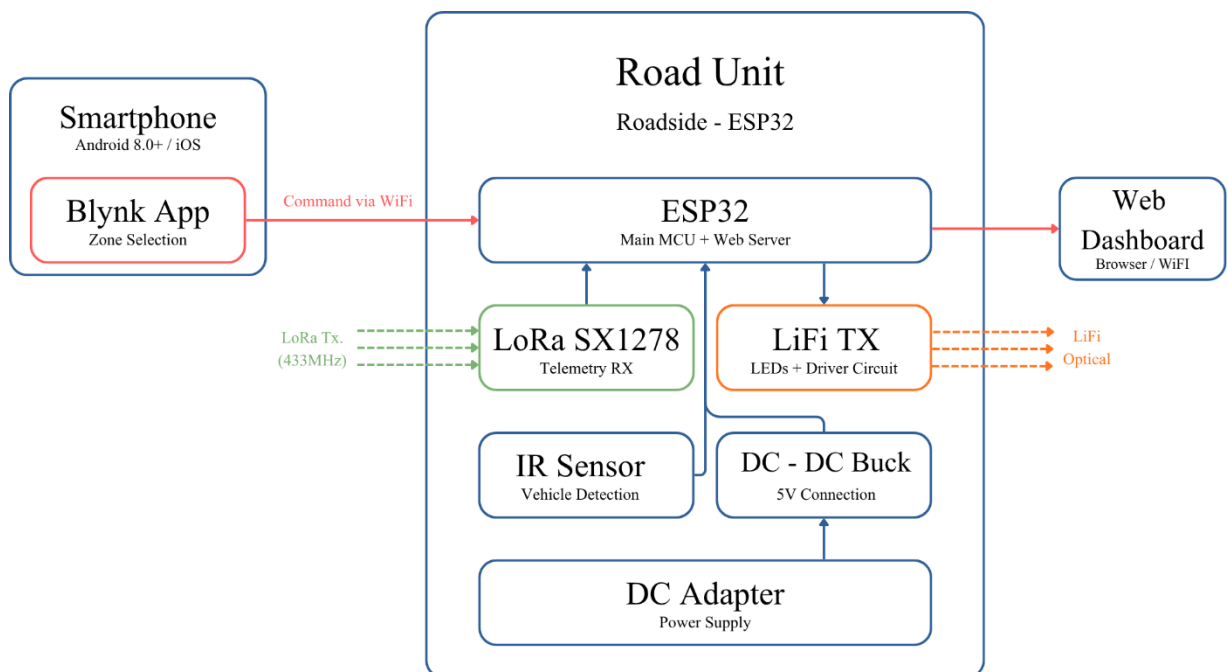


Figure 5.2: Proposed System Architecture Diagram – Road Unit

5.3 WORKFLOW OF THE PROPOSED SYSTEM

Figures 5.3 and 5.4 represent the workflow of the proposed framework associated with vehicular speed regulation. The workflow of the proposed system begins with the initialisation of the roadside control unit, where the predefined speed limit for the specific restricted zone is configured in the RU firmware [4]. Once powered, the RU continuously monitors its IR sensor output. When a vehicle enters the detection zone, the IR sensor triggers the microcontroller, which immediately activates LiFi transmission — encoding the speed limit value into a modulated LED signal using ON/OFF binary encoding with start, data, and stop bits.

On the vehicle side, the VU's photodiode continuously samples incoming light. When the modulated LiFi signal is received, the microcontroller decodes the signal, extracts the speed limit value, and compares it with the current vehicle speed. If the vehicle speed exceeds the received limit, the PWM duty cycle is progressively reduced to slow the motor smoothly.

Simultaneously, the LoRa module transmits the current speed and system status to the monitoring station at regular intervals. If the driver activates the override button, speed limiting is suspended for 20 seconds before automatic control resumes. If the LiFi signal is lost, the system maintains the last valid speed or defaults to a safe fallback speed [14].

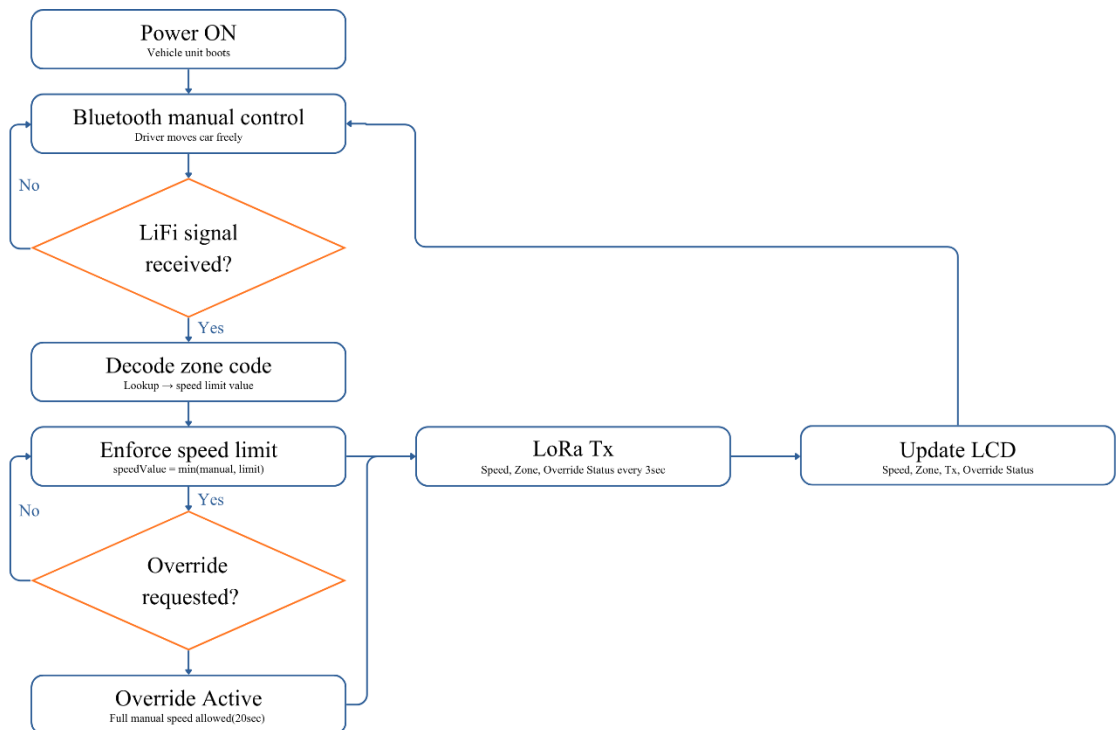


Figure 5.3: Workflow Diagram of the Proposed System – Vehicle Unit

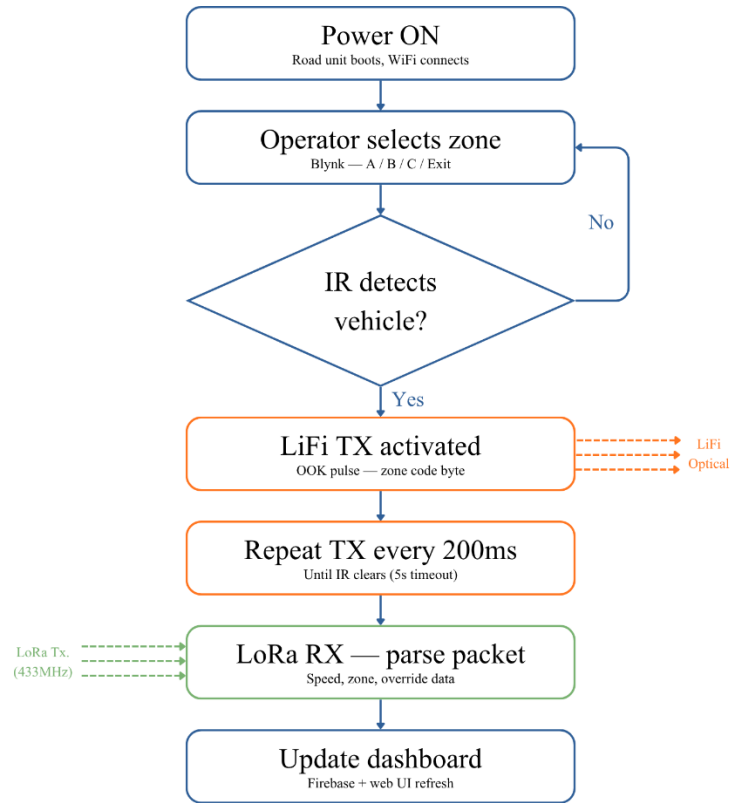


Figure 5.4: Workflow Diagram of the Proposed System – Road Unit

5.4 LIST OF MODULES

The proposed hybrid LiFi–LoRa based intelligent vehicle speed control system is divided into the following functional modules:

- i. **Vehicle Detection Module (IR Sensor)** - Detects approaching vehicles using an IR sensor and activates the LiFi transmission system only when a vehicle is present.
- ii. **Speed Limit Transmission Module (LiFi)** - Transmits the predefined speed limit information from the roadside unit to the vehicle using LiFi-based LED modulation.
- iii. **Signal Reception and Decoding Module** - Receives the LiFi signal through a photodiode, decodes the transmitted speed limit data, and validates the received information.
- iv. **PWM Motor Speed Control Module** - Controls and regulates vehicle motor speed using PWM signals based on the decoded speed limit.
- v. **LoRa Telemetry Module** - Sends real-time vehicle telemetry and system status data to the remote monitoring station using LoRa communication.
- vi. **Driver Alert and Override Module** - Provides visual/audible alerts to the driver and enables temporary manual override during emergency situations.

- vii. **Fail-Safe Module** - Ensures safe operation during LiFi signal loss by maintaining the last valid speed limit or switching to a predefined safe speed.

5.5 MODULE DESCRIPTION

Module 1: Vehicle Detection Module (IR Sensor)

This module forms the first layer of the hybrid communication architecture. The IR sensor mounted on the roadside unit continuously monitors the road for approaching vehicles. When a vehicle enters the sensor's detection range, the sensor output changes state, triggering the microcontroller to activate the LiFi transmission module. This ensures that the LiFi LED is only active when a vehicle is actually present, conserving energy and preventing unnecessary signal broadcast.

Module 2: Speed Limit Transmission Module (LiFi)

Upon receiving the trigger from the IR detection module, the roadside microcontroller encodes the configured speed limit value into a binary digital signal and drives the high-brightness LED using ON/OFF modulation. The modulation scheme uses a defined protocol with start bits, data bits (representing the speed limit), and stop bits to ensure reliable decoding at the vehicle end. The LiFi signal is broadcast within a defined spatial cone covering the entry point of the restricted zone.

Module 3: Signal Reception and Decoding Module

The photodiode mounted on the vehicle VU captures the modulated light signal from the roadside LED. The ESP32 microcontroller samples the photodiode output at a frequency matched to the LiFi modulation rate, reconstructs the binary data stream, and decodes the speed limit value. Error handling logic is implemented to discard corrupted frames and retain the last valid decoded value. An optional amplifier circuit between the photodiode and the microcontroller enhances signal amplitude and noise immunity.

Module 4: PWM Motor Speed Control Module

Once the speed limit value is decoded, the VU microcontroller compares it with the current motor speed — inferred from the PWM duty cycle or an optional encoder — and adjusts the PWM output to the L298N motor driver accordingly. Speed reduction is applied gradually using a ramp-down algorithm to ensure smooth deceleration and passenger comfort. When the vehicle exits the restricted zone and no further LiFi signal is received, the system restores the motor to its pre-limited speed.

Module 5: LoRa Telemetry Module

The LoRa SX1278 module on the VU continuously transmits telemetry packets at configurable intervals to the remote monitoring station. Each packet contains the current vehicle speed (as a PWM duty cycle value), the system status (normal / speed-limited / override active), and a timestamp. The monitoring station, equipped with a corresponding LoRa receiver module connected to a PC or cloud gateway, logs and displays this data in real time, enabling remote supervision of the vehicle's compliance with speed restrictions.

Module 6: Driver Alert and Override Module

Visual or audible alerts — using an LED indicator or buzzer — notify the driver when speed limiting is actively enforced. The manual override button allows the driver to temporarily suspend speed limiting for 20 seconds in emergency situations. The override event is logged and transmitted via LoRa to the monitoring station. After the 20-second window, the system automatically returns to autonomous speed control mode.

Module 7: Fail-Safe Module

If the LiFi signal is not received for a defined timeout period — due to optical obstruction, misalignment, or sensor malfunction — the fail-safe module either maintains the last successfully decoded speed limit or switches the vehicle to a predefined safe default speed. This ensures that a signal loss event does not result in the vehicle resuming unrestricted speed in a zone that may still be restricted. The fail-safe status is also reported via LoRa to the monitoring station.

CHAPTER 6

IMPLEMENTATION

6.1 SYSTEM IMPLEMENTATION

The system implementation involves two independently powered and programmed embedded units — the Road Unit (RU) and the Vehicle Unit (VU) — that communicate through the hybrid IR-LiFi-LoRa channel. Both units are programmed using Embedded C within the Arduino framework, developed and compiled using Visual Studio Code with the PlatformIO extension ^[11].

The RU firmware initialises the IR sensor interface and configures the LED GPIO pin for LiFi modulation output. It continuously polls the IR sensor and, upon detection, executes the LiFi transmission subroutine to broadcast the configured speed limit. The VU firmware initialises the LEDC PWM peripheral for motor control, configures the SPI interface for the LoRa module, and runs the photodiode sampling and signal decoding routines concurrently through interrupt-driven or timer-based architecture.

6.2 WORKING OF EACH MODULE

6.2.1 IR Detection and LiFi Transmission (Road Unit)

The RU microcontroller continuously reads the digital output of the IR sensor. When the output transitions — indicating a vehicle in the detection zone — an interrupt or polling event triggers the LiFi transmission function. The speed limit value (e.g., 30 km/h encoded as binary 00011110) is serialised into a bit stream prefixed with a start bit and terminated with a stop bit ^[15]. The microcontroller drives the LED GPIO high and low according to the bit stream at a defined baud rate, creating a detectable modulated optical signal.

6.2.2 LiFi Signal Reception and Decoding (Vehicle Unit)

The VU microcontroller samples the photodiode output at a rate equal to or greater than the LiFi baud rate. The sampled values are processed to detect the start bit, after which the subsequent data bits are captured and assembled into the speed limit byte. A parity check or frame validation step ensures only correctly formatted frames are accepted ^[12]. The decoded speed limit value is stored in a global variable accessible to the motor control subroutine. Invalid or corrupted frames are discarded, and the last valid value is retained.

6.2.3 PWM Motor Speed Control

The ESP32's LEDC (LED Control) peripheral is configured to generate PWM signals at a frequency suitable for the L298N motor driver ^[11]. The duty cycle, ranging from 0% to 100%, directly controls the motor speed. When a speed limit is received, the control algorithm computes the target duty cycle corresponding to the received limit and gradually ramps the current duty cycle down to the target over a defined time period. This ramp-down prevents abrupt speed changes that could destabilise the vehicle or cause discomfort ^[4].

6.2.4 LoRa Telemetry Transmission

At regular intervals (configurable, e.g., every 500 ms), the VU packages the current PWM duty cycle (as a speed indicator), the active system status flag (0 = normal, 1 = speed-limited, 2 = override), and a sequential packet ID into a LoRa data frame. The SX1278 module transmits this frame at the configured spreading factor and frequency ^[21]. The monitoring station's LoRa receiver captures and parses the frame, logging the data and displaying it on a connected terminal or cloud dashboard ^[3].

6.2.5 Manual Override and Fail-Safe

When the override button is pressed, an interrupt handler sets the override flag, bypasses the speed limiting logic, and starts a 20-second countdown timer. During the override period, the LoRa telemetry frame includes the override status flag. When the timer expires, the override flag is cleared, and the system resumes normal autonomous speed control. For the fail-safe, a LiFi signal watchdog timer monitors the time since the last valid frame reception ^[14]. If this exceeds the threshold, the system applies the safe default speed and sets the fail-safe status flag in the LoRa telemetry.

6.3 SNAPSHOTS

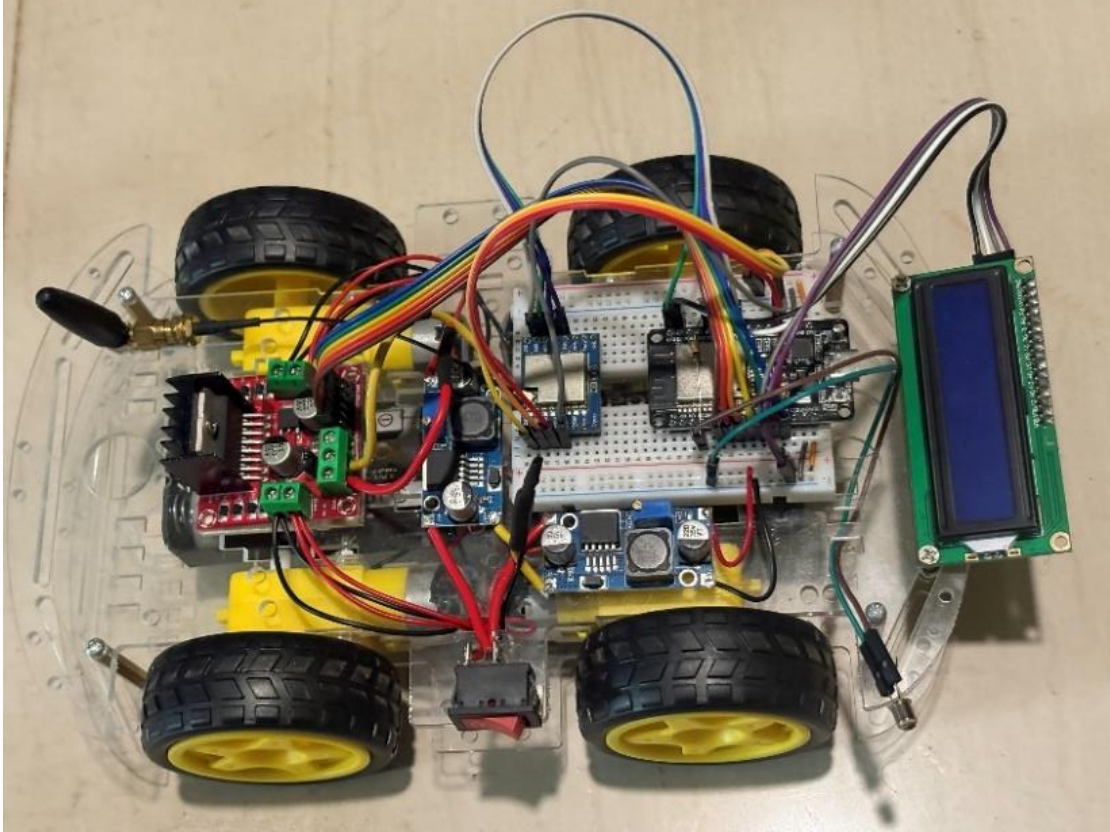


Figure 6.1: Vehicle Unit Circuit Setup

In Figure 6.1, the Vehicle Unit circuit setup is presented, mounted on a four-wheeled RC car chassis. The setup integrates the ESP32 microcontroller as the main processing unit, connected to the L298N motor driver for controlling the four DC drive motors. A breadboard is used for component interconnections, with the LiFi phototransistor receiver mounted to capture incoming optical signals from the roadside unit. The 16x2 I2C LCD display is mounted on the side of the chassis, providing real-time feedback of vehicle speed, zone status, transmission state, and override status during operation.

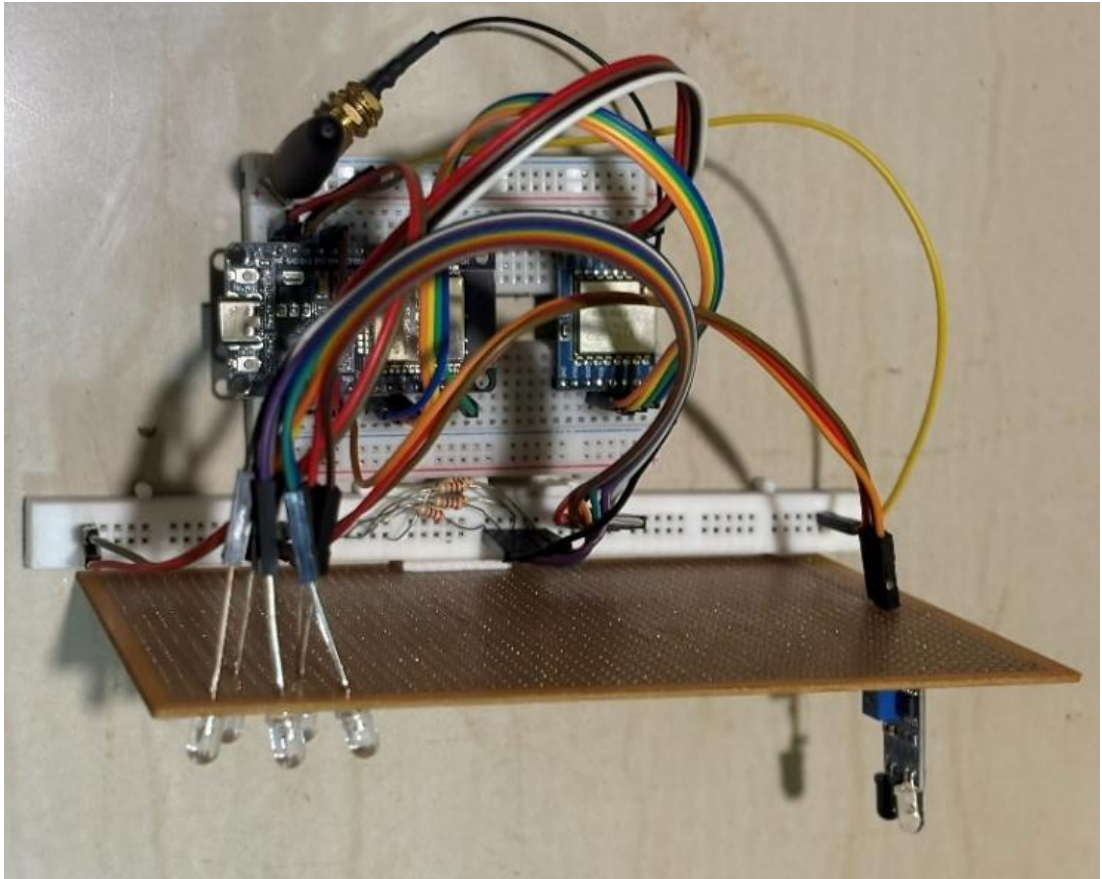


Figure 6.2: Road Unit Circuit Setup

In Figure 6.2, the Road Unit circuit setup is presented, mounted on a flat wooden platform simulating a roadside installation. The setup integrates the ESP32 microcontroller as the main processing unit, connected to the LiFi LED array — consisting of multiple high-brightness LEDs — mounted at the front of the platform for optical speed limit transmission. The LoRa SX1278 module is mounted on the breadboard with its antenna extended for telemetry reception from the vehicle unit. The IR sensor is positioned at the base of the platform to detect approaching vehicles and trigger the LiFi transmission sequence.

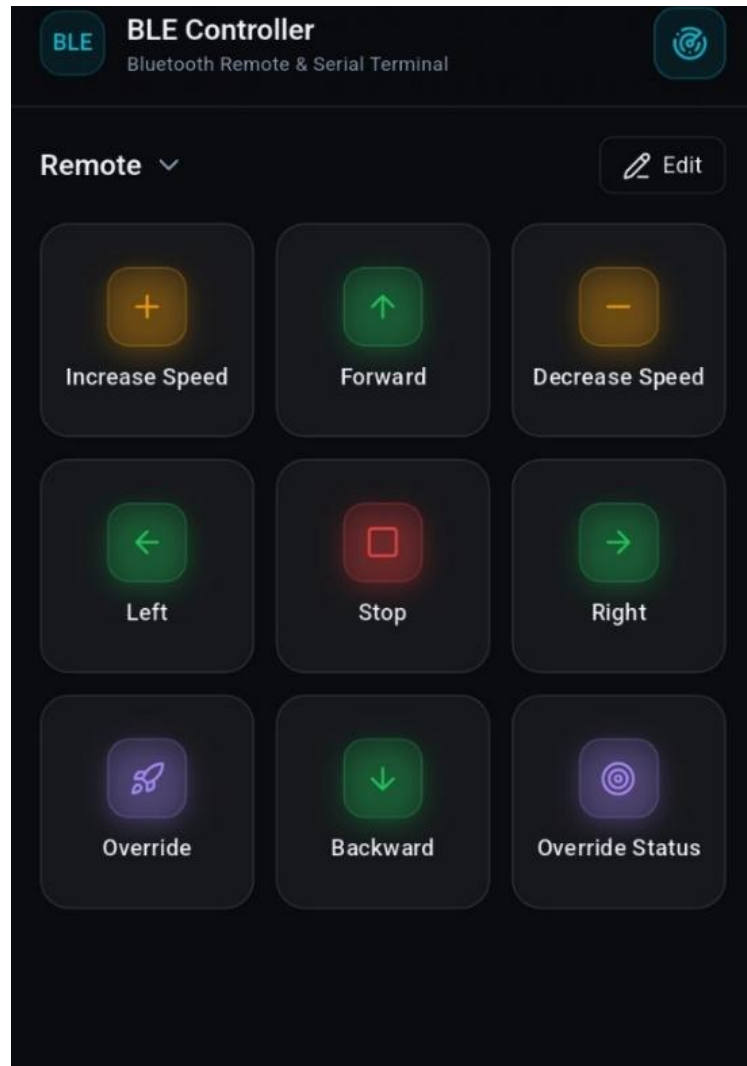


Figure 6.3: Vehicle Unit Bluetooth Controller Application – BLE Controller

In Figure 6.3, the Bluetooth Controller mobile application interface used to manually control the Vehicle Unit is presented. The application — running on an Android smartphone — communicates with the ESP32 via Bluetooth Serial, sending single-character command strings for vehicle movement and speed control. The interface provides dedicated buttons for Forward, Backward, Left, Right, Stop, Increase Speed, and Decrease Speed operations. Additionally, an Override button allows the driver to activate the manual override mechanism from the application, and an Override Status button toggles the LCD status screen on the vehicle unit for real-time override and cooldown information.

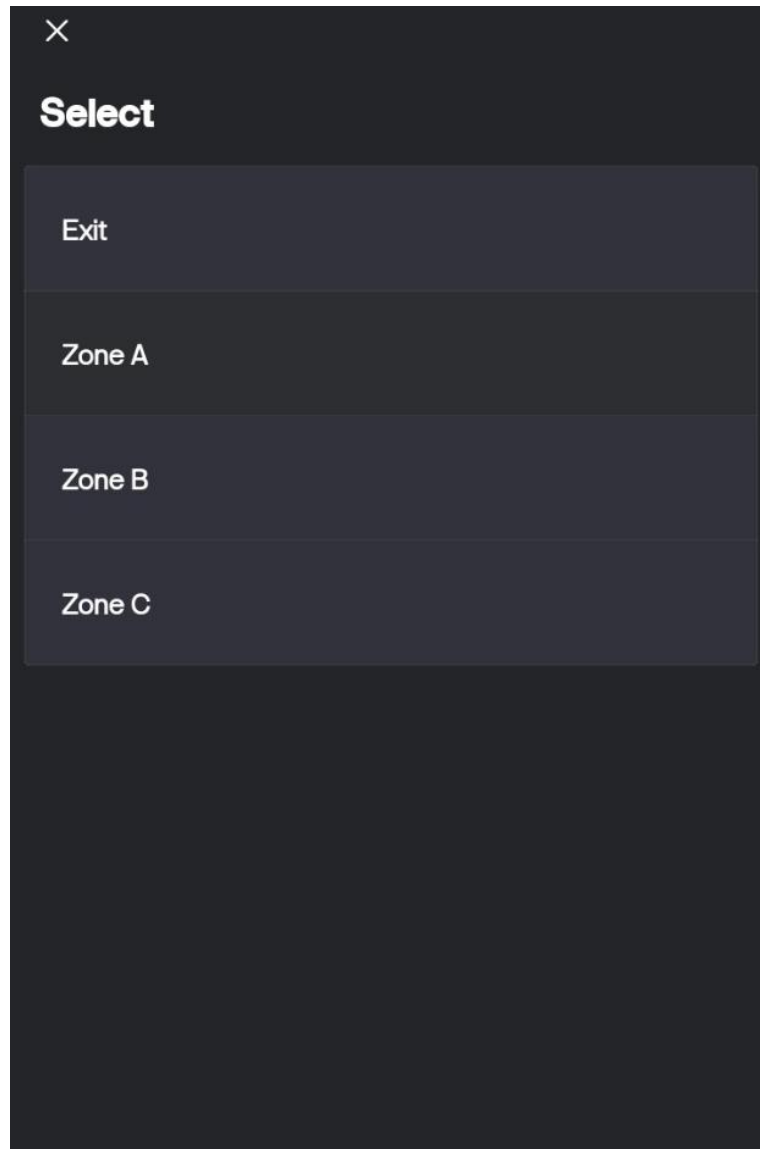


Figure 6.4: Road Unit Zone Selection Application – Blynk App

In Figure 6.4, the Blynk mobile application interface used to configure the Road Unit's active speed zone is presented. The application communicates with the ESP32 Road Unit over Wi-Fi using the Blynk IoT platform. The interface provides a simple dropdown selector with four options — Exit, Zone A, Zone B, and Zone C — each corresponding to a different speed limit value encoded and transmitted via the LiFi channel. Selecting a zone instantly updates the zone code on the Road Unit, which then begins broadcasting the corresponding speed limit upon vehicle detection by the IR sensor.

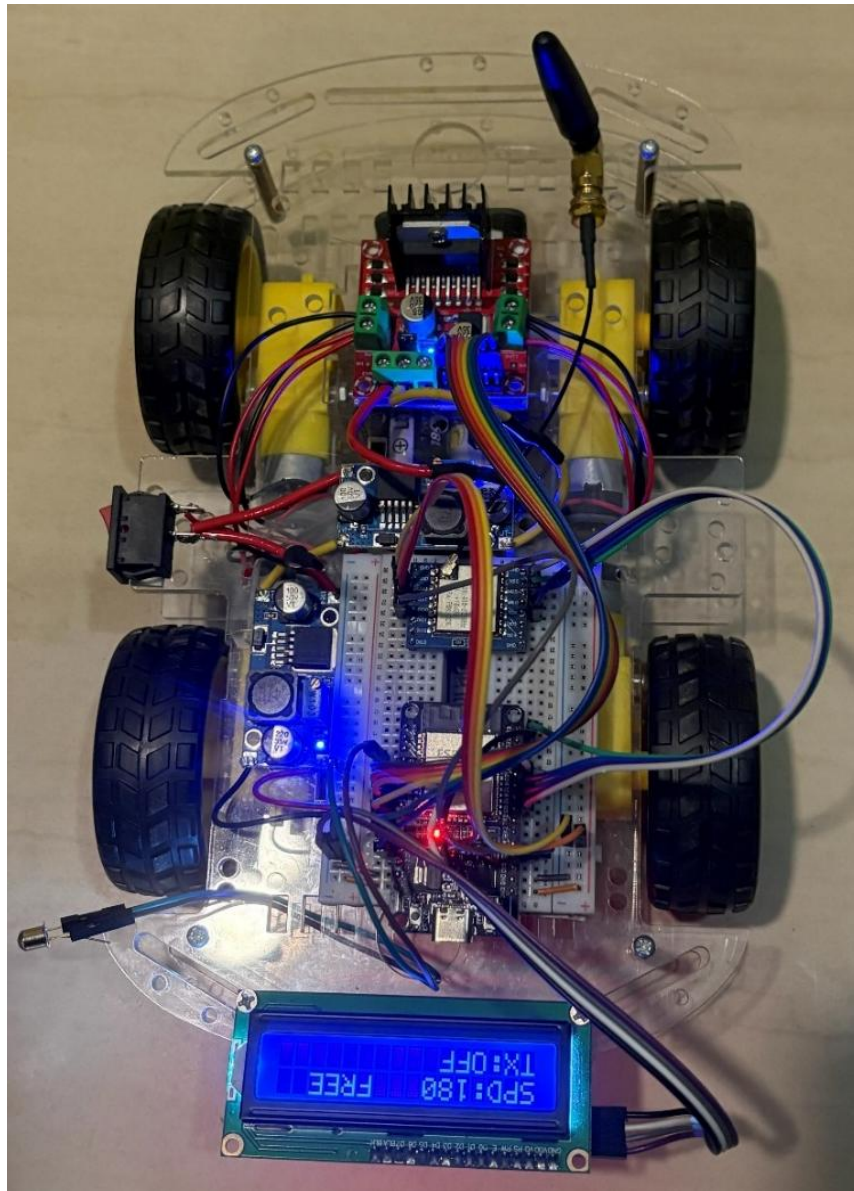


Figure 6.5: Vehicle Unit – With Manual Control

In Figure 6.5, the Vehicle Unit is shown in operation under manual Bluetooth control, with the 16x2 I2C LCD display clearly visible at the front of the chassis. The LCD display reads "SPD:180" on the first line and "TX:OFF FREE" on the second line, indicating that the vehicle is currently operating at a PWM speed value of 180 in free-zone mode with no active LiFi zone detected and no LoRa transmission in progress. This snapshot demonstrates the baseline manual operation state of the vehicle unit prior to entering a restricted speed zone.

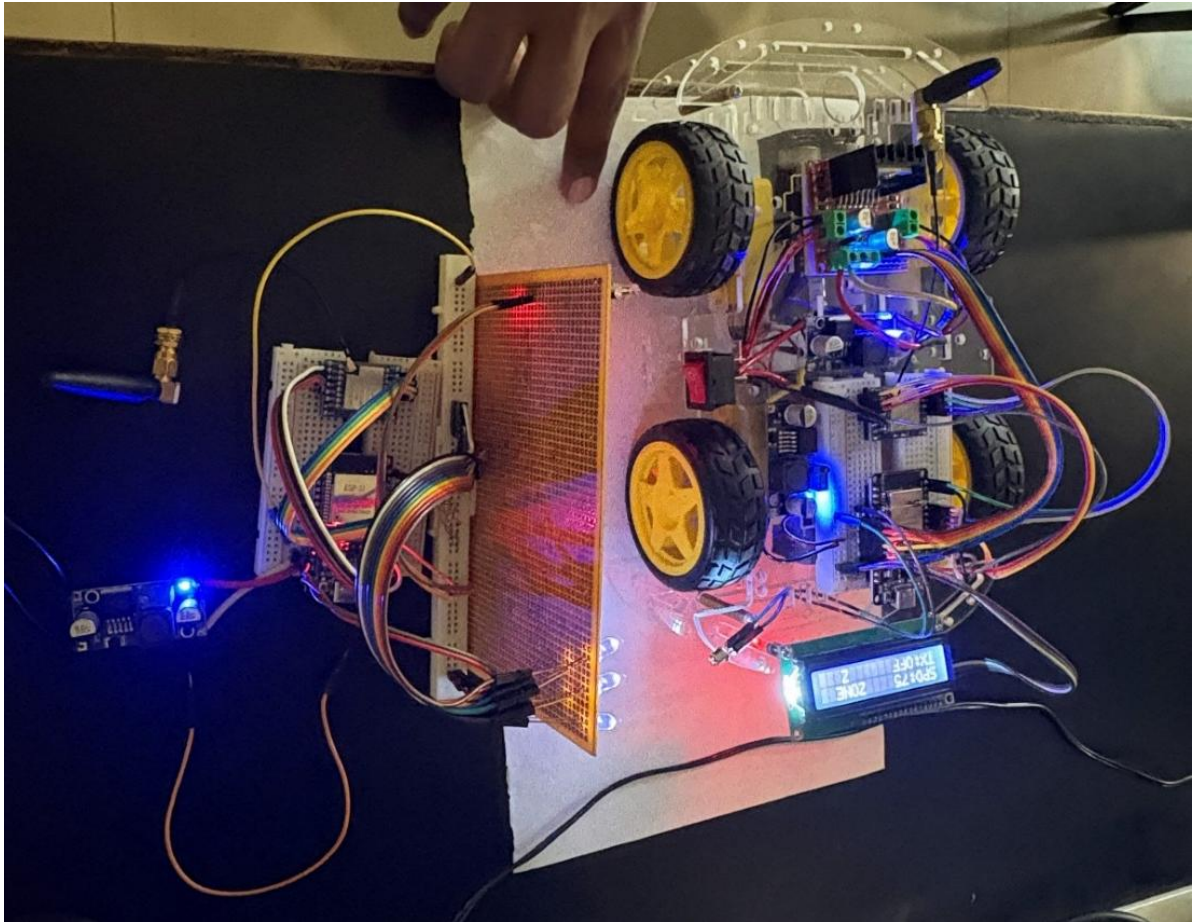


Figure 6.6: LiFi Transmission and Reception

In Figure 6.6, the active LiFi transmission and reception between the Road Unit and the Vehicle Unit is demonstrated. The Road Unit's high-brightness LEDs are visibly illuminated — emitting a modulated optical signal carrying the speed limit zone code — directed toward the Vehicle Unit's phototransistor receiver. The blue indicator LED on the Vehicle Unit confirms active signal reception. The LCD display on the Vehicle Unit reflects the updated system state upon successful LiFi decoding, with the zone status and enforced speed limit displayed in real time. This snapshot validates the core optical communication channel of the proposed hybrid system.

CHAPTER 7

RESULTS AND DISCUSSION

The proposed Intelligent Vehicle Speed Limiting and Monitoring System was implemented as a prototype-level robotic vehicle setup and tested under controlled indoor conditions to evaluate the performance of each communication and control module. The tests focused on verifying the reliability of IR-based vehicle detection, the accuracy of LiFi-based speed limit transmission and decoding, the responsiveness of PWM-based motor speed control, and the reliability of LoRa telemetry over varying distances ^[3].

7.1 LiFi Communication Performance

LiFi signal transmission and reception was tested at distances ranging from 0.1 m to 1 m under controlled ambient light conditions. The system successfully decoded the speed limit value with high accuracy within this range. The directional nature of the LiFi channel ensured that only the vehicle directly in front of the roadside LED received the signal, with no unintended reception at oblique angles beyond the LED's emission cone ^[14]. Frame error rates were minimal at standard operating distances.

7.2 Motor Speed Control Response

Upon receiving the speed limit signal, the PWM motor control algorithm smoothly ramped down the motor speed to the target limit within the configured ramp period. The vehicle exhibited controlled and gradual deceleration without abrupt stops or instability ^[4]. When the vehicle exited the restricted zone and LiFi signals ceased, the motor speed was restored to the normal operating level. The override mechanism successfully suspended speed limiting for exactly 20 seconds in all test scenarios.

7.3 LoRa Telemetry Performance

LoRa telemetry was tested at distances ranging from 10 m to over 200 m in an open indoor/outdoor environment. Packet delivery was consistent, with telemetry updates received at the monitoring station at the configured interval ^[2]. The system status (normal, speed-limited, override) was accurately reflected in each transmitted packet, enabling reliable remote supervision of the vehicle's state throughout the test runs ^[3].

Table 7.1: Performance Analysis

Test Parameter	Metric	Result	Observation
LiFi Decoding Accuracy	Frame Error Rate	< 2% at 1 m	Reliable within rated range
IR Detection Reliability	Detection Rate	100% at 0-10 cm	Consistent trigger on vehicle entry
PWM Speed Ramp Response	Time to Target Speed	< 500 ms	Smooth and gradual deceleration
Override Timer Accuracy	Override Duration	20 sec $\pm$ 0.1 sec	Precise timer-based control
LoRa Packet Delivery	Packet Success Rate	> 90% at 200 m	Reliable telemetry over test range
Fail-Safe Activation	Timeout Detection	< 1 sec	Correct fallback to safe speed

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENT

This project successfully demonstrated the design and implementation of an Intelligent Vehicle Speed Limiting and Monitoring System using a hybrid IR-LiFi-LoRa communication architecture. By combining Infrared detection for vehicle presence, LiFi optical communication for real-time speed limit transmission, and LoRa wireless telemetry for remote monitoring, the system delivers an autonomous, cost-effective, and reliable alternative to existing speed regulation approaches ^{[3][12][17]}.

Unlike conventional systems that depend on driver compliance or post-event enforcement, the proposed system proactively enforces speed limits at the vehicle level without requiring cellular network infrastructure or high-cost dedicated hardware. The PWM-based motor speed control ensures smooth and safe deceleration, the override mechanism accommodates emergency scenarios, and the fail-safe logic ensures system reliability even under signal loss conditions ^[4]. The prototype-level implementation on a robotic RC car setup validated the core functionality of all seven system modules under controlled test conditions.

The system directly addresses the limitations identified in the literature and the existing system analysis — namely the lack of real-time vehicle-level speed enforcement, high infrastructure costs, and driver dependence ^{[8][16]}— while aligning with the sustainable development goals of good health, innovation, and sustainable cities.

Overall Future Enhancements

- **GPS-Based Zone Verification:** Integration of GPS location data to cross-verify that the vehicle is within the designated restricted zone before applying speed limits, reducing false triggering from non-vehicular objects ^[7].
- **Encrypted LiFi Communication:** Implementation of lightweight encryption on the LiFi data channel to prevent spoofing attacks where unauthorised sources attempt to inject false speed limits ^[12].
- **Vehicle CAN Bus Integration:** Extension of the system beyond prototype RC car to real vehicle platforms via CAN bus interface for direct throttle and braking control.
- **Multi-Zone Scalability:** Deployment of multiple RUs across a route, each broadcasting different speed limits, with the VU seamlessly transitioning between zones.

- Machine Learning for Adaptive Speed Control: Use of ML algorithms to adapt speed limits dynamically based on traffic density, time of day, and environmental conditions, moving beyond static preset limits ^[4].
- Cloud-Based Fleet Monitoring: Scalable cloud integration enabling simultaneous monitoring of multiple vehicles across multiple restricted zones, with historical data analytics and violation reporting ^[11].

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APPENDIX - I

SOURCE CODE

IOT_ROAD_UNIT:

platform.ini

```
[env:esp32dev]
platform = espressif32
board = esp32dev
framework = arduino
monitor_speed = 115200
lib_deps =
    blynkkk/Blynk
    sandeepmistry/LoRa
```

main.cpp

```
// INTELLIGENT VEHICLE SPEED LIMITING SYSTEM
// Road Unit — IR + LiFi TX + LoRa RX + Web Dashboard
// Team: Dharshnamoorthy R, Gogulakannan M,
//       Henry Daniel Didace D, Prashanth Arjun GK
// MVIT, Pondicherry University | 2025-26
// PIN SUMMARY:
//   LoRa SX1278 : SS=5, RST=14, DIO0=2
//               SCK=18, MISO=19, MOSI=23
//   IR Sensor   : GPIO 4
//   LiFi TX     : GPIO 13 (BC547 base via 1kΩ)
//   WiFi       : Mobile Hotspot (Blynk + Web Server)
// BLYNK VIRTUAL PINS:
//   V0 — Zone selector (0=Exit, 1=A, 2=B, 3=C)
// ----- BLYNK -----
#define BLYNK_TEMPLATE_ID "TMPL3WGA3ObfN"
#define BLYNK_TEMPLATE_NAME "Road Unit"
#define BLYNK_AUTH_TOKEN "ROpTOj95yRzj6UN1nL01mnsQoWp19Cy2"
#include <Arduino.h>
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <SPI.h>
#include <LoRa.h>
#include <WebServer.h>
// WIFI CREDENTIALS
char ssid[] = "Embedded";
char pass[] = "12348765";
// LORA PINS
#define LORA_SS 5
#define LORA_RST 14
#define LORA_DIO0 2
// IR + LIFI PINS
#define IR_PIN 4
#define LIFI_PIN 13
// LIFI PULSE ENCODING
// OOK (On-Off Keying) pulse encoding
// Bit 1 = 500µs HIGH + 500µs LOW
// Bit 0 = 200µs HIGH + 800µs LOW
// Start = 1000µs HIGH + 500µs LOW
#define LIFI_BIT1_HIGH 500
#define LIFI_BIT1_LOW 500
#define LIFI_BIT0_HIGH 200
#define LIFI_BIT0_LOW 800
#define LIFI_START_HIGH 1000
#define LIFI_START_LOW 500
#define LIFI_REPEAT_MS 200 // Repeat packet every 200ms
// ZONE DEFINITIONS
// Zone codes transmitted via LiFi (1 byte each)
```

```

// Vehicle lookup table must match these values
#define ZONE_EXIT 0x00
#define ZONE_A 0xA1 // Speed limit 75
#define ZONE_B 0xB2 // Speed limit 102
#define ZONE_C 0xC3 // Speed limit 128
// ZONE STATE
int selectedZone = 0; // 0=Exit, 1=A, 2=B, 3=C (from Blynk)
byte zoneCode = ZONE_EXIT;
int zoneLimitValue = 0; // For display purposes
bool vehicleDetected = false;
bool lifiActive = false;

// IR DETECTION
unsigned long irDetectTime = 0;
const unsigned long IR_HOLD = 5000; // Keep LiFi TX on for 5s after detection
// LIFI TX TIMER
unsigned long lastLiFiTx = 0;
// LORA RX — LATEST DATA
String loraZone = "---";
int loraSpeed = 0;
int loraLimit = 0;
bool loraOverride = false;
int loraOvrCount = 0;
String loraLastMsg = "---";
unsigned long loraLastRx = 0;
// WEB SERVER
WebServer server(80);
// DASHBOARD HTML
// Auto-refreshes every 3 seconds
String buildDashboard() {
  String uptime = String(millis() / 1000) + "s";
  String zoneLabel = "---";
  if (selectedZone == 1) zoneLabel = "Zone A";
  else if (selectedZone == 2) zoneLabel = "Zone B";
  else if (selectedZone == 3) zoneLabel = "Zone C";
  else
    zoneLabel = "No Zone";
  String vehicleStatus = vehicleDetected ? "DETECTED" : "CLEAR";
  String lifiStatus = lifiActive ? "ACTIVE" : "IDLE";
  String ovrStatus = loraOverride ? "YES" : "NO";
  // Time since last LoRa packet
  String lastSeen = "---";
  if (loraLastRx > 0) {
    unsigned long ago = (millis() - loraLastRx) / 1000;
    lastSeen = String(ago) + "s ago";
  }
  String html = R"rawhtml(
<!DOCTYPE html>
<html>
<head>
  <meta charset='UTF-8'>
  <meta name='viewport' content='width=device-width, initial-scale=1'>
  <meta http-equiv='refresh' content='3'>
  <title>Speed Limit Monitor</title>
  <style>
    * { box-sizing: border-box; margin: 0; padding: 0; }
    body {
      font-family: 'Segoe UI', sans-serif;
      background: #0f1117;
      color: #e0e0e0;
      padding: 20px;
    }
    h1 {
      text-align: center;

```

```

font-size: 1.4rem;
color: #ffffff;
margin-bottom: 6px;
letter-spacing: 1px;
}
.subtitle {
text-align: center;
font-size: 0.75rem;
color: #555;
margin-bottom: 20px;
}
.grid {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(160px, 1fr));
gap: 14px;
max-width: 700px;
margin: 0 auto 20px auto;
}
.card {
background: #1a1d27;
border: 1px solid #2a2d3e;
border-radius: 10px;
padding: 16px 12px;
text-align: center;
}
.card .label {
font-size: 0.7rem;
color: #666;
text-transform: uppercase;
letter-spacing: 1px;
margin-bottom: 8px;
}
.card .value {
font-size: 1.6rem;
font-weight: 700;
color: #ffffff;
}
.card .value.green { color: #4caf82; }
.card .value.red { color: #e05252; }
.card .value.amber { color: #f0a500; }
.card .value.blue { color: #4a9eff; }
.section {
max-width: 700px;
margin: 0 auto 14px auto;
background: #1a1d27;
border: 1px solid #2a2d3e;
border-radius: 10px;
padding: 14px 16px;
}
.section h2 {
font-size: 0.75rem;
color: #555;
text-transform: uppercase;
letter-spacing: 1px;
margin-bottom: 10px;
}
.row {
display: flex;
justify-content: space-between;
padding: 5px 0;
border-bottom: 1px solid #22253a;
font-size: 0.88rem;
}

```

```

.row:last-child { border-bottom: none; }
.row .k { color: #888; }
.row .v { color: #ddd; font-weight: 600; }
.badge {
  display: inline-block;
  padding: 2px 10px;
  border-radius: 20px;
  font-size: 0.75rem;
  font-weight: 700;
}
.badge.green { background: #1a3a2a; color: #4caf82; }
.badge.red { background: #3a1a1a; color: #e05252; }
.badge.amber { background: #3a2a0a; color: #f0a500; }
.footer {
  text-align: center;
  font-size: 0.7rem;
  color: #333;
  margin-top: 10px;
}
</style>
</head>
<body>
<h1>&#128246; Speed Zone Monitor</h1>
<p class='subtitle'>MVIT — Intelligent Vehicle Speed Limiting System</p>
<div class='grid'>
  <div class='card'>
    <div class='label'>Vehicle Speed</div>
    <div class='value blue'><rawhtml">
html += loraSpeed;
html += R"rawhtml(</div>
    </div>
  <div class='card'>
    <div class='label'>Zone Limit</div>
    <div class='value amber'><rawhtml">
html += (loraLimit > 0 ? String(loraLimit) : "---");
html += R"rawhtml(</div>
    </div>
  <div class='card'>
    <div class='label'>Zone Status</div>
    <div class='value'><rawhtml">
html += (loraZone == "IN" ? "green" : "red");
html += ">";
html += loraZone;
html += R"rawhtml(</div>
    </div>
  <div class='card'>
    <div class='label'>Override</div>
    <div class='value'><rawhtml">
html += (loraOverride ? "red" : "green");
html += ">";
html += ovrStatus;
html += R"rawhtml(</div>
    </div>
  </div>
<div class='section'>
  <h2>Road Unit Status</h2>
  <div class='row'>
    <span class='k'>Active Zone</span>
    <span class='v'><rawhtml">
html += zoneLabel;
html += R"rawhtml(</span>
  </div>
  <div class='row'>

```

```

        <span class='k'>IR Detection</span>
        <span class='v'>)rawhtml";
html += "<span class='badge ' + String(vehicleDetected ? "green" : "red") + ">" + vehicleStatus + "</span>";
html += R"rawhtml(</span>
    </div>
    <div class='row'>
        <span class='k'>LiFi TX</span>
        <span class='v'>)rawhtml";
html += "<span class='badge ' + String(lifiActive ? "green" : "amber") + ">" + lifiStatus + "</span>";
html += R"rawhtml(</span>
    </div>
</div>

<div class='section'>
    <h2>LoRa Telemetry</h2>
    <div class='row'>
        <span class='k'>Last Packet</span>
        <span class='v'>)rawhtml";
html += lastSeen;
html += R"rawhtml(</span>
    </div>
    <div class='row'>
        <span class='k'>Override Count</span>
        <span class='v'>)rawhtml";
html += loraOvrCount;
html += R"rawhtml( / 5</span>
    </div>
    <div class='row'>
        <span class='k'>Raw Message</span>
        <span class='v' style='font-size:0.75rem;color:#666'>)rawhtml";
html += loraLastMsg;
html += R"rawhtml(</span>
    </div>
</div>
<div class='footer'>Auto-refresh every 3s &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;Uptime: )rawhtml";
html += uptime;
html += R"rawhtml( &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp; )rawhtml";
html += WiFi.localIP().toString();
html += R"rawhtml(</div>
</body>
</html>
)rawhtml";
return html;
}
// WEB SERVER HANDLERS
void handleRoot() {
    server.send(200, "text/html", buildDashboard());
}
void handleNotFound() {
    server.send(404, "text/plain", "Not found");
}
// LIFI — TRANSMIT ONE BIT
void lifiSendBit(bool bit) {
    if (bit) {
        digitalWrite(LIFI_PIN, HIGH);
        delayMicroseconds(LIFI_BIT1_HIGH);
        digitalWrite(LIFI_PIN, LOW);
        delayMicroseconds(LIFI_BIT1_LOW);
    } else {
        digitalWrite(LIFI_PIN, HIGH);
        delayMicroseconds(LIFI_BIT0_HIGH);
        digitalWrite(LIFI_PIN, LOW);
        delayMicroseconds(LIFI_BIT0_LOW);
    }
}

```

```

    }
}
// LIFI — TRANSMIT ONE BYTE
// MSB first, 8 bits
void lifiSendByte(byte data) {
    // Start pulse
    digitalWrite(LIFI_PIN, HIGH);
    delayMicroseconds(LIFI_START_HIGH);
    digitalWrite(LIFI_PIN, LOW);
    delayMicroseconds(LIFI_START_LOW);
    // 8 data bits MSB first
    for (int i = 7; i >= 0; i--) {
        lifiSendBit((data >> i) & 0x01);
    }
    // Stop — LED off
    digitalWrite(LIFI_PIN, LOW);
}
// LIFI — TRANSMIT ZONE CODE
// Only fires if zone selected AND vehicle detected
void handleLiFiTx() {
    // Conditions check
    if (selectedZone == 0) {
        lifiActive = false;
        digitalWrite(LIFI_PIN, LOW);
        return;
    }
    if (!vehicleDetected) {
        lifiActive = false;
        digitalWrite(LIFI_PIN, LOW);
        return;
    }
    // Repeat every LIFI_REPEAT_MS
    if (millis() - lastLiFiTx < LIFI_REPEAT_MS) return;
    lastLiFiTx = millis();
    lifiActive = true;
    lifiSendByte(zoneCode);
    Serial.print("[LIFI] TX Zone code: 0x");
    Serial.println(zoneCode, HEX);
}
// IR SENSOR — VEHICLE DETECTION
// Active LOW (most IR modules pull LOW on detect)
void handleIR() {
    if (digitalRead(IR_PIN) == LOW) {
        irDetectTime = millis();
        if (!vehicleDetected) {
            vehicleDetected = true;
            Serial.println("[IR] Vehicle detected");
        }
    }
}
// Hold detection for IR_HOLD ms after last trigger
if (vehicleDetected &&
    millis() - irDetectTime > IR_HOLD) {
    vehicleDetected = false;
    lifiActive = false;
    digitalWrite(LIFI_PIN, LOW);
    Serial.println("[IR] Vehicle cleared");
}
}
// LORA RX — PARSE INCOMING PACKET
// Expected format:
// ZONE:IN,SPD:75,LIM:75,OVR:0,OVRC:0
// ZONE:OUT,SPD:180,OVR:0

```

```

void parseLoRaPacket(String msg) {
  loraLastMsg = msg;
  loraLastRx = millis();
  // Helper lambda-style inline parser
  auto extract = [&](String key) -> String {
    int idx = msg.indexOf(key + ":");
    if (idx == -1) return "---";
    int start = idx + key.length() + 1;
    int end = msg.indexOf(" ", start);
    if (end == -1) end = msg.length();
    return msg.substring(start, end);
  };
  loraZone = extract("ZONE");
  loraSpeed = extract("SPD").toInt();
  loraLimit = extract("LIM").toInt();
  loraOverride = (extract("OVR") == "1");
  loraOvrCount = extract("OVRC").toInt();
  Serial.println("[LORA] RX: ");
  Serial.println(msg);
}
// LORA RX — CHECK FOR INCOMING PACKET
void handleLoRaRx() {
  int packetSize = LoRa.parsePacket();
  if (packetSize == 0) return;
  String msg = "";
  while (LoRa.available()) {
    msg += (char)LoRa.read();
  }
  parseLoRaPacket(msg);
}
// BLYNK — ZONE SELECTOR
// V0: 0=Exit, 1=Zone A, 2=Zone B, 3=Zone C
BLYNK_WRITE(V0) {
  selectedZone = param.asInt();
  switch (selectedZone) {
    case 1:
      zoneCode = ZONE_A;
      zoneLimitValue = 75;
      Serial.println("[BLYNK] Zone A selected — LIM 75");
      break;
    case 2:
      zoneCode = ZONE_B;
      zoneLimitValue = 102;
      Serial.println("[BLYNK] Zone B selected — LIM 102");
      break;
    case 3:
      zoneCode = ZONE_C;
      zoneLimitValue = 128;
      Serial.println("[BLYNK] Zone C selected — LIM 128");
      break;
    default:
      zoneCode = ZONE_EXIT;
      zoneLimitValue = 0;
      lifiActive = false;
      digitalWrite(LIFI_PIN, LOW);
      Serial.println("[BLYNK] Zone EXIT — LiFi OFF");
      break;
  }
}
// SETUP
void setup() {
  Serial.begin(115200);
  // ----- PINS -----

```

```

pinMode(IR_PIN, INPUT_PULLUP);
pinMode(LIFI_PIN, OUTPUT);
digitalWrite(LIFI_PIN, LOW);
// ----- WIFI -----
Serial.print("[WIFI] Connecting");
WiFi.begin(ssid, pass);
int retries = 0;
while (WiFi.status() != WL_CONNECTED && retries < 20) {
    delay(500);
    Serial.print(".");
    retries++;
}

if (WiFi.status() == WL_CONNECTED) {
    Serial.println("\n[WIFI] Connected");
    Serial.print("[WIFI] IP: ");
    Serial.println(WiFi.localIP());
} else {
    Serial.println("\n[WIFI] Failed — continuing without WiFi");
}
// ----- BLYNK -----
Blynk.config(BLYNK_AUTH_TOKEN);
Blynk.connect();
// ----- WEB SERVER -----
server.on("/", handleRoot);
server.onNotFound(handleNotFound);
server.begin();
Serial.println("[WEB] Server started");
Serial.print("[WEB] Dashboard: http://");
Serial.println(WiFi.localIP());
// ----- LORA -----
LoRa.setPins(LORA_SS, LORA_RST, LORA_DIO0);
if (!LoRa.begin(433E6)) {
    Serial.println("[LORA] INIT FAILED");
    while (1);
}
LoRa.setSpreadingFactor(7);
LoRa.setSignalBandwidth(125E3);
LoRa.setCodingRate4(5);
Serial.println("[LORA] Ready at 433 MHz");
Serial.println("[SYS] Road unit online");
}
// LOOP
void loop() {
    Blynk.run(); // Blynk zone selection
    server.handleClient(); // Web dashboard
    handleIR(); // IR vehicle detection
    handleLiFiTx(); // LiFi zone code transmission
    handleLoRaRx(); // LoRa telemetry reception
}

```

IOT_VEHICLE_UNIT:

platform.ini

```

[env:esp32dev]
platform = espressif32
board = esp32dev
framework = arduino
monitor_speed = 115200
lib_deps =
    marcoschwartz/LiquidCrystal_I2C
    sandeepmistry/LoRa

```

main.cpp

```

// INTELLIGENT VEHICLE SPEED LIMITING SYSTEM
// Vehicle Unit — Bluetooth RC Car
// Team: Dharshnamoorthy R, Gogulakannan M,
//      Henry Daniel Didace D, Prashanth Arjun GK
// MVIT, Pondicherry University | 2025-26
#include <Arduino.h>
#include "BluetoothSerial.h"
#include <SPI.h>
#include <LoRa.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
BluetoothSerial SerialBT;
// LCD
LiquidCrystal_I2C lcd(0x27, 16, 2);
String lastLCDLine0 = "";
String lastLCDLine1 = "";
// MOTOR PINS (L298N)
#define ENA 32
#define IN1 33
#define IN2 25
#define ENB 12
#define IN3 26
#define IN4 27
// LIFI PIN
#define LIFI_PIN 34
// LORA PINS (SPI)
#define LORA_SS 5
#define LORA_RST 14
#define LORA_DIO0 2
// SPEED SETTINGS
int manualSpeed = 180; // Set by Bluetooth U/D commands
int speedValue = 180; // Actual applied speed
int zoneSpeed = 75; // Speed limit set by LiFi zone
// MOVEMENT FLAGS
bool fwd = false;
bool back = false;
bool left = false;
bool right = false;
// ZONE & TX STATE
bool inZone = false;
bool txActive = false;
// LIFI TIMER
unsigned long lastLiFiTime = 0;
const unsigned long LIFI_TIMEOUT = 3000; // 3s signal loss = exit zone
// LORA TIMER
unsigned long lastLoRaSend = 0;
const unsigned long LORA_INTERVAL = 3000; // Send every 3s
// OVERRIDE SYSTEM
bool overrideActive = false;
bool overrideCooldownActive = false;
bool overrideLocked = false;
unsigned long overrideStart = 0;
unsigned long overrideCooldownStart = 0;
unsigned long overrideLockStart = 0;
const unsigned long OVERRIDE_DURATION = 20000; // 20 sec active
const unsigned long OVERRIDE_COOLDOWN = 100000; // 100 sec cooldown
const unsigned long OVERRIDE_LOCKTIME = 1200000; // 20 min lock
int overrideCount = 0;
const int OVERRIDE_MAX = 5;
// LCD SCREEN TOGGLE
bool showStatusScreen = false;
// LCD UPDATE — NO FLICKER (no lcd.clear())
// Writes only changed lines

```

```

void updateLCD() {
    String line0 = "";
    String line1 = "";
    // ----- STATUS SCREEN -----
    if (showStatusScreen) {
        line0 = "OVR:";
        line0 += overrideCount;
        line0 += "/";
        line0 += OVERRIDE_MAX;
        line0 += " ";
        line0 += overrideActive ? "ACT " : "IDLE";
        if (overrideCooldownActive) {
            unsigned long remain =
                (OVERRIDE_COOLDOWN -
                 (millis() - overrideCooldownStart)) / 1000;
            line1 = "CD:";
            line1 += remain;
            line1 += "s ";
        }
        else if (overrideLocked) {
            unsigned long remain =
                (OVERRIDE_LOCKTIME -
                 (millis() - overrideLockStart)) / 60000;
            line1 = "LOCK:";
            line1 += remain;
            line1 += "m ";
        }
        else {
            line1 = "READY ";
        }
    }
    // ----- NORMAL SCREEN -----
    else {
        line0 = "SPD:";
        line0 += speedValue;
        line0 += " ";
        // Right-align zone indicator at col 10
        line0 = line0.substring(0, 10);
        line0 += inZone ? "ZONE" : "FREE";
        line0 += " ";
        line1 = "TX:";
        line1 += txActive ? "ON " : "OFF ";
        line1 += " ";
        line1 = line1.substring(0, 8);
        line1 += overrideActive ? "OVR " : " ";
        line1 += inZone ? "Z" : " ";
    }
    // Only write if changed — eliminates flicker
    if (line0 != lastLCDLine0) {
        lcd.setCursor(0, 0);
        lcd.print(line0.substring(0, 16));
        lastLCDLine0 = line0;
    }
    if (line1 != lastLCDLine1) {
        lcd.setCursor(0, 1);
        lcd.print(line1.substring(0, 16));
        lastLCDLine1 = line1;
    }
}
// MOTOR CONTROL
void moveCar() {
    int leftSpeed = speedValue;
    int rightSpeed = speedValue;

```

```

int in1 = LOW, in2 = LOW;
int in3 = LOW, in4 = LOW;
// ----- FORWARD -----
if (fwd && !back) {
    in1 = HIGH; in2 = LOW;
    in3 = HIGH; in4 = LOW;
}
// ----- BACKWARD -----
else if (back && !fwd) {
    in1 = LOW; in2 = HIGH;
    in3 = LOW; in4 = HIGH;
}
// ----- TURN WHILE MOVING -----
if (left && (fwd || back)) {
    leftSpeed /= 2;
}
else if (right && (fwd || back)) {
    rightSpeed /= 2;
}
// ----- SPOT TURN (no fwd/back) -----
else if (left && !fwd && !back) {
    // Left motor reverse, right motor forward = spot left
    in1 = LOW; in2 = HIGH;
    in3 = HIGH; in4 = LOW;
}
else if (right && !fwd && !back) {
    // Left motor forward, right motor reverse = spot right
    in1 = HIGH; in2 = LOW;
    in3 = LOW; in4 = HIGH;
}
// ----- FULL STOP -----
if (!fwd && !back && !left && !right) {
    leftSpeed = 0;
    rightSpeed = 0;
}
digitalWrite(IN1, in1);
digitalWrite(IN2, in2);
digitalWrite(IN3, in3);
digitalWrite(IN4, in4);
ledcWrite(0, leftSpeed);
ledcWrite(1, rightSpeed);
updateLCD();
}
// APPLY SPEED LOGIC
// Priority: Override > Zone > Manual
void applySpeedLogic() {
    if (overrideActive) {
        speedValue = manualSpeed;    // Override: use manual
    }
    else if (inZone) {
        speedValue = min(manualSpeed, zoneSpeed); // Zone: enforce limit
        if (manualSpeed > zoneSpeed) {
            Serial.print("[SPD] Capped to zone limit: ");
            Serial.println(zoneSpeed);
        }
    }
    else {
        speedValue = manualSpeed;    // Free: use manual
    }
    moveCar();
}
// OVERRIDE — ACTIVATE
void activateOverride() {

```

```

if (!inZone) {
    Serial.println("[OVR] Only inside zone");
    return;
}
if (overrideLocked) {
    Serial.println("[OVR] LOCKED");
    return;
}
if (overrideCooldownActive) {
    Serial.println("[OVR] Cooldown active");
    return;
}
overrideActive = true;
overrideStart = millis();
overrideCount++;

Serial.print("[OVR] ACTIVATED (");
Serial.print(overrideCount);
Serial.print(" ");
Serial.print(OVERRIDE_MAX);
Serial.println(")");
applySpeedLogic();
if (overrideCount >= OVERRIDE_MAX) {
    overrideLocked = true;
    overrideLockStart = millis();
    Serial.println("[OVR] SYSTEM LOCKED — 20 min");
}
}
// OVERRIDE — TIMER HANDLER (non-blocking)
void handleOverride() {
    // Override duration expired
    if (overrideActive &&
        millis() - overrideStart >= OVERRIDE_DURATION) {
        overrideActive = false;
        overrideCooldownActive = true;
        overrideCooldownStart = millis();
        Serial.println("[OVR] Ended — Cooldown started");
        applySpeedLogic();
    }
    // Cooldown expired
    if (overrideCooldownActive &&
        millis() - overrideCooldownStart >= OVERRIDE_COOLDOWN) {
        overrideCooldownActive = false;
        Serial.println("[OVR] Cooldown cleared");
    }
    // Lock expired
    if (overrideLocked &&
        millis() - overrideLockStart >= OVERRIDE_LOCKTIME) {
        overrideLocked = false;
        overrideCount = 0;
        Serial.println("[OVR] Lock cleared");
    }
}
// BLUETOOTH — NON-BLOCKING READ
void handleBluetooth() {
    if (!SerialBT.available()) return;
    String cmd = "";
    unsigned long timeout = millis();
    // Read until newline or 50ms timeout
    while (millis() - timeout < 50) {
        if (SerialBT.available()) {
            char c = SerialBT.read();
            if (c == '\n' || c == '\r') break;

```

```

    cmd += c;
}
}
cmd.trim();
if (cmd.length() == 0) return;
Serial.println("[BT] CMD: " + cmd);
// ---- MOVEMENT ----
if (cmd == "F") { fwd = true; back = false; }
else if (cmd == "B") { back = true; fwd = false; }
else if (cmd == "L") { left = true; }
else if (cmd == "R") { right = true; }
else if (cmd == "S") {
    fwd = false; back = false;
    left = false; right = false;
}

// ---- SPEED UP ----
else if (cmd == "U") {
    manualSpeed += 5;
    if (manualSpeed > 255) manualSpeed = 255;
    Serial.print("[SPD] Manual: "); Serial.println(manualSpeed);
    applySpeedLogic();
    return;
}
// ---- SPEED DOWN ----
else if (cmd == "D") {
    manualSpeed -= 5;
    if (manualSpeed < 0) manualSpeed = 0;
    Serial.print("[SPD] Manual: "); Serial.println(manualSpeed);
    applySpeedLogic();
    return;
}

// ---- OVERRIDE ----
else if (cmd == "O") {
    activateOverride();
    return;
}
// ---- STATUS SCREEN TOGGLE ----
else if (cmd == "ST") {
    showStatusScreen = !showStatusScreen;
    updateLCD();
    return;
}
else {
    Serial.println("[BT] Unknown cmd: " + cmd);
    return;
}
moveCar();
}
// LIFI ZONE DETECTION
// Current progress: LOW signal = inside zone
void checkLiFiZone() {
    // Ignore LiFi for first 2 seconds after boot
    if (millis() < 2000) return;
    if (digitalRead(LIFI_PIN) == LOW) {
        lastLiFiTime = millis();
        if (!inZone) {
            inZone = true;
            Serial.println("[LIFI] ENTER ZONE");
            applySpeedLogic();
        }
    }
}
}

```

```

// Signal lost for > LIFI_TIMEOUT → exit zone
if (inZone && millis() - lastLiFiTime > LIFI_TIMEOUT) {
  inZone = false;
  overrideActive = false;
  Serial.println("[LIFI] EXIT ZONE");
  applySpeedLogic();
}
}

// LORA TRANSMIT
// Sends inside zone (status) + outside zone (heartbeat)
// No blocking delay — non-blocking implementation
void sendLoRa() {

  if (millis() - lastLoRaSend < LORA_INTERVAL) return;
  lastLoRaSend = millis();
  txActive = true;
  updateLCD();
  String msg = "";
  if (inZone) {
    msg += "ZONE:IN";
    msg += ",SPD:";
    msg += speedValue;
    msg += ",LIM:";
    msg += zoneSpeed;
    msg += ",OVR:";
    msg += (overrideActive ? "1" : "0");
    msg += ",OVRC:";
    msg += overrideCount;
  }
  else {
    // Heartbeat — let monitoring station know car is alive
    msg += "ZONE:OUT";
    msg += ",SPD:";
    msg += speedValue;
    msg += ",OVR:0";
  }

  LoRa.beginPacket();
  LoRa.print(msg);
  LoRa.endPacket();
  Serial.println("[LORA] TX: " + msg);
  txActive = false;
  updateLCD();
}

// SETUP
void setup() {
  Serial.begin(115200);
  // ----- BLUETOOTH -----
  SerialBT.begin("ESP32_RC_CAR");
  Serial.println("[BT] Started: ESP32_RC_CAR");
  // ----- LCD -----
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("SYSTEM INIT...");
  delay(1000);
  lcd.clear();
  lastLCDLine0 = "";
  lastLCDLine1 = "";
  // ----- MOTOR PINS -----
  pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT);

```

```

pinMode(IN3, OUTPUT); pinMode(IN4, OUTPUT);
// ----- LIFI -----
pinMode(LIFI_PIN, INPUT);
// ----- PWM (LEDC) -----
ledcSetup(0, 1000, 8);
ledcAttachPin(ENA, 0);
ledcSetup(1, 1000, 8);
ledcAttachPin(ENB, 1);
// ----- LORA -----
LoRa.setPins(LORA_SS, LORA_RST, LORA_DIO0);
if (!LoRa.begin(433E6)) {
  Serial.println("[LORA] INIT FAILED");
  lcd.setCursor(0, 0);
  lcd.print("LORA FAIL!   ");
  while (1);
}
LoRa.setSpreadingFactor(7); // SF7 — fast, short range
LoRa.setSignalBandwidth(125E3); // 125 kHz BW
LoRa.setCodingRate4(5); // CR 4/5
Serial.println("[LORA] Ready at 433 MHz");
// ----- READY -----
updateLCD();
Serial.println("[SYS] Vehicle unit online");
}
// LOOP
void loop() {
  handleBluetooth(); // BT commands — non-blocking
  //checkLiFiZone(); // LiFi zone detection
  handleOverride(); // Override timer management
  sendLoRa(); // LoRa telemetry — non-blocking
}

```

APPENDIX - II
LIST OF PUBLICATION